

ChatGPT Conversation: Capitalizing Development Costs

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Turn 1: User

Time: 2026-05-01 15:07:59 GMT+8

Example 1: development costs under IFRS v s US GAAP Under IAS 38:

- research costs are expensed when incurred
- development costs are capitalized if specific criteria are met

Under the SEC's comparison of IFRS and US GAAP:

- research and development costs are generally expensed as incurred under US GAAP
- under IFRS, qualifying development costs can be capitalized

Why this matters:

- an IFRS software or pharma company can show higher assets and higher current-period

earnings than a similar US GAAP company

- later periods then pick u p amortization
- so ROA, EBITDA-style metrics, asset growth, and margin comparisons can all be distorted

i f you treat the statements as directly comparable

can you explain how the capitalizing the resesrch works? do you mean you can borrow long term loan to pay for research? or do you use cash? wouldnt that reduce earnings though? or capex is not subtracted from earnings usually?

Turn 2: Assistant

Time: 2026-05-01 15:08:01 GMT+8

Good question — you're mixing two different ideas: how something is financed (cash vs loan) vs how it's accounted for (expense vs capitalized asset). The IFRS vs US GAAP difference is about the second, not the first.

Let's break it cleanly.

1) What “capitalizing development costs” actually means

Under IAS 38:

- Research phase → expense immediately
- **Development phase → capitalize *if criteria are met*** (technical feasibility, intent to complete, future economic benefit, etc.)

Capitalizing = putting it on the balance sheet as an asset

Instead of this:

Expense → reduces profit immediately

You do this:

Asset (intangible) → spread cost over future periods via amortization

2) Simple numerical example

Case A — Expense everything (US GAAP style)

Say you spend \$100 on development:

- Income statement:
 - Expense = 100
 - Profit ↓ by 100
 - Balance sheet:
 - No asset created
-

Case B — Capitalize (IFRS development phase)

- Year 1:
 - Expense = 0 (or minimal)
 - Profit does NOT drop much
 - Balance sheet: +100 intangible asset
- Future years:

- Amortization (say 20/year for 5 years)
 - Profit ↓ gradually over time
-

3) Does this depend on cash vs loan?

No — completely separate.

You can fund development with:

- Cash
- Debt (loan)
- Equity

Example:

- Borrow \$100
- Spend on development

Accounting entries:

- Liability: +100 (loan)
- Asset: +100 (capitalized development)

Cash is just the mechanism. Capitalization is about classification, not funding.

4) Why doesn't capex reduce earnings immediately?

Because of the core accounting principle:

Matching principle → match costs with the revenue they generate

Development is assumed to create future benefits, so IFRS says:

- Don't hit earnings all at once
 - Spread the cost over time (amortization)
-

5) Key intuition

Think of it like building a factory:

- You don't expense the entire factory cost in one year
- You depreciate it over time

IFRS treats certain development costs like a "software factory"

6) Why this distorts comparisons

Under US GAAP:

- Most R&D → expensed immediately

Under IFRS:

- Some development → capitalized

Result:

An IFRS company may show:

- Higher profits (initially)
 - Higher assets
 - Higher ROA denominator later
 - Lower future earnings (due to amortization)
-

7) Your specific questions answered directly

do you mean you can borrow long term loan to pay for research?

You can, but that's unrelated. Capitalization is not about borrowing — it's about how you record the cost.

or do you use cash?

Yes, usually you pay salaries, engineers, etc. in cash — but again, accounting treatment is independent of this.

wouldn't that reduce earnings though?

- If expensed → YES immediately
 - If capitalized → NO (initially), YES later via amortization
-

capex is not subtracted from earnings usually?

Correct.

- Capex (capitalized cost) → goes to balance sheet
 - Only depreciation/amortization hits earnings
-

8) Mental model (best way to remember)

- Expense = "this benefits only today"

- Capitalize = “this creates future value”

IFRS just allows more things (like development) to be treated as the second category.

If you want, I can walk through a real software company example (like how this shows up in financials) — that’s usually where it really clicks.

Turn 3: User

Time: 2026-05-01 15:15:38 GMT+8

when paying for r&d with cash, it would lower cash balance and increase intangible assets, so net assets will remain unchanged?

can you distinguish IAS, US GAAP, and IFRS? where are they from as well?

Turn 4: Assistant

Time: 2026-05-01 15:15:39 GMT+8

You’re very close — just one key nuance to fix.

1) Does paying cash for R&D keep net assets unchanged?

[OK] If the cost is capitalized (IFRS development phase)

Then your intuition is correct:

- Cash ↓ 100
- Intangible asset ↑ 100

Net assets = unchanged (at that moment)

-> You’ve just converted one asset (cash) into another (intangible)

☐ But over time...

That asset gets amortized:

- Expense recognized later
- Retained earnings ↓
- Net assets ↓ gradually

So:

- Day 1 → no change
- Future → net assets decline via amortization

☐ If the cost is expensed (research or US GAAP R&D)

- Cash ↓ 100
- Expense = 100 → profit ↓ → retained earnings ↓

Net assets ↓ immediately

☐ Summary table

Treatment	Cash	Asset	Earnings	Net assets
Capitalize	↓	↑	no immediate hit	same (initially)
Expense	↓	none	↓ immediately	↓ immediately

2) IAS vs IFRS vs US GAAP (who they are)

This part confuses a lot of people — here’s the clean mental model.

☐☐ US GAAP

- Who sets it: Financial Accounting Standards Board
- Where used: United States
- Nature: Rules-based (more detailed, prescriptive)

-> Requires most R&D to be expensed immediately

☐ IFRS

- Who sets it: International Accounting Standards Board
- Where used: Most of the world (Europe, Korea, etc.)
- Nature: Principles-based (more judgment)

-> Allows development costs to be capitalized if criteria met

☐ IAS

- These are individual standards within IFRS

Example:

- IAS 38 → governs R&D accounting
- IAS 16 → for physical assets

Key relationship

Think of it like this:

- IFRS = the whole system
- IAS = specific rulebooks inside IFRS

Analogy:

IFRS = operating system
IAS = individual apps

☐ Historical note (why both names exist)

- Before 2001 → standards were called IAS
- After 2001 → new standards called IFRS
- Old IAS standards (like IAS 38) are still in use

3) One clean mental model

- US GAAP: conservative → expense early
- IFRS: smoother earnings → capitalize when justified

4) Final intuition (ties everything together)

When you capitalize development:

- You're saying:

"This spending is not gone — it's stored as future value"

When you expense:

- You're saying:

"This value is uncertain — treat it as gone today"
--

If you want, I can show how this flows through cash flow statement vs income statement vs balance sheet together — that's where this really clicks for investors.

Turn 5: User

Time: 2026-05-01 16:30:09 GMT+8

A model trained on post-impairment recoveries under IFRS could misread those earnings rebounds as operating improvement rather than accounting reversal mechanics.

what does this mean? can you give examples of impairments as well?

Turn 6: Assistant

Time: 2026-05-01 16:30:10 GMT+8

This sentence is basically warning about a pattern-recognition mistake:

A model (or investor) might think "profits went up → business improved,"
when in reality the increase is just accounting reversing a past loss, not better operations.

Let's unpack it step by step.

1) What is an impairment?

Under IFRS (specifically IAS 36):

An impairment happens when:

The value of an asset on the books is higher than what it's actually worth (recoverable amount)

So the company must:

- Write it down
 - Record a loss in earnings
-

☐ Example of impairment

A company builds software:

- Capitalized development cost: 100
- But product fails → expected future cash flow drops

Now the asset is only worth 40

-> Impairment:

- Asset: 100 → 40
 - Loss: 60 hits income statement
-

Other real-world examples

1) Pharma company

- Spends years developing a drug
- Trial fails

→ Entire project may be impaired (huge write-off)

2) Tech company (like your domain)

- Capitalized software project
- Users don't adopt it

→ Write down the capitalized dev cost

3) Goodwill impairment

- Company overpays for acquisition
- Expected synergies don't materialize

→ Goodwill impairment (often very large)

2) What is a “post-impairment recovery”?

Here's the key IFRS twist:

Under IFRS (unlike US GAAP):

Some impairments can be reversed if conditions improve later

☐ Example of reversal

Continue previous example:

- After impairment: asset = 40

- Later, product unexpectedly succeeds
- New value = 80

-> IFRS allows:

- Asset: 40 → 80
- Gain: +40 in income statement

3) Why this confuses models (or investors)

Now imagine the income statement over time:

Year	Event	Profit impact
-----	-----	-----
Year 1	Impairment	-60
Year 2	Reversal	+40

A model sees:

“Profit jumped +40 → business improved!”

But reality:

- No new operational improvement necessarily
- Just accounting undoing part of a previous loss

4) What the sentence means (plain English)

“A model trained on post-impairment recoveries under IFRS could misread those earnings rebounds as operating improvement rather than accounting reversal mechanics.”

Translation:

- The model sees earnings going up
- It assumes:

“Company is doing better”

But actually:

- The increase is from:

“We previously wrote the asset down too much, now we’re adjusting it back up”

5) Why this matters (especially for you building systems)

This creates false signals:

- EBITDA ↑
- Net income ↑
- ROA changes

But:

- Cash flow may not change
 - Core business may be unchanged
-

6) IFRS vs US GAAP difference (important)

Under US GAAP:

- Impairments are generally NOT reversed

So:

- No “fake rebound” effect

Under IFRS:

- Reversals allowed (except goodwill)

-> That's why cross-country comparisons get messy

7) Clean mental model

- Impairment = “we were too optimistic before”
- Reversal = “we were too pessimistic before”

Neither necessarily means:

“the business suddenly improved right now”

8) Quick analogy

Think of it like correcting a mistake:

- Year 1:

“Oops, I overstated value → subtract 60”

- Year 2:

“Oops again, I overcorrected → add back 40”

A model might think:

“Wow, performance improved in Year 2!”

But really:

It's just accounting corrections bouncing around

If you want, I can show how this shows up in real company financial statements (e.g., tech or pharma cases) — that's where this becomes very concrete.

Turn 7: User

Time: 2026-05-01 16:36:15 GMT+8

what is roa

Turn 8: Assistant

Time: 2026-05-01 16:36:16 GMT+8

ROA (Return on Assets) is a basic profitability ratio that tells you:

How efficiently a company uses its assets to generate profit

The formula

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total Assets}}$$

1) Intuition

- Assets = everything the company owns (cash, equipment, software, etc.)
- Net income = profit after all expenses

-> ROA answers:

“For every \$1 of assets, how much profit is the company generating?”

2) Simple example

- Net income = 10
- Total assets = 100

$$\text{ROA} = 10 / 100 = 10\%$$

-> Meaning:

- The company generates 10 cents of profit per \$1 of assets
-

3) Why it matters

ROA helps you compare:

- Efficiency across companies
- Capital intensity of businesses

Examples:

- Software company → usually high ROA (few physical assets)
 - Manufacturing → usually lower ROA (heavy machinery, factories)
-

4) Tie-back to your earlier question (very important)

ROA gets distorted by IFRS vs US GAAP differences:

Under IFRS

- Development costs can be capitalized
- → Assets ↑
- → Net income initially ↑

Under US GAAP

- R&D mostly expensed
- → Assets lower
- → Net income lower initially

-> So ROA comparisons can be misleading:

- IFRS company might look more profitable or less efficient, depending on timing
-

5) One subtle but important nuance

ROA can move for two very different reasons:

1. Real improvement
 - Better margins
 - More revenue
2. Accounting effects

- Capitalizing development
- Impairments or reversals

-> That's why analysts often adjust ROA to make companies comparable

6) Quick mental model

- High ROA → "This company squeezes a lot of profit out of its assets"
 - Low ROA → "This business needs a lot of assets to generate profit"
-

If you want, I can show how ROA changes step-by-step when you capitalize vs expense development costs — that's a really powerful intuition builder.

Turn 9: User

Time: 2026-05-01 16:41:23 GMT+8

does goodwill impairment get recorded as assets? and does it get amortized?

Turn 10: Assistant

Time: 2026-05-01 16:41:24 GMT+8

Short answer: No and no — goodwill impairment is not recorded as an asset, and goodwill is not amortized (under IFRS and US GAAP).

Let's unpack it cleanly.

1) What goodwill is (baseline)

Goodwill arises when you acquire a company for more than the fair value of its net assets.

- Example:
 - You buy a company for 150
 - Its identifiable net assets = 100
 - -> Goodwill = 50 (recorded as an intangible asset)
-

2) What happens when goodwill is impaired?

Under both:

- IFRS
- US GAAP

-> If goodwill is impaired:

- Asset (goodwill) ↓
- Expense (impairment loss) hits income statement

Example:

- Goodwill on balance sheet = 50
- New estimated value = 20

-> Impairment:

- Goodwill: 50 → 20
- Loss: 30 (reduces earnings)

☐ Key point

The impairment itself is not an asset

It is a loss (expense)

So:

- You are reducing an asset, not creating one
-

3) Is goodwill amortized?

☐ No (under IFRS and US GAAP)

Instead:

- Goodwill is:
 - Tested annually for impairment
 - Written down if needed
-

Why not amortized?

Because:

Goodwill has an indefinite life

Examples:

- Brand reputation
- Customer relationships
- Synergies

These don't "expire" in a predictable schedule

4) Important IFRS vs US GAAP nuance

Under IFRS

- Goodwill impairment cannot be reversed

Under US GAAP

- Also cannot be reversed

-> So goodwill is special:

- Other assets → sometimes reversible under IFRS
 - Goodwill → never reversed
-

5) Contrast with other intangible assets

Type	Amortized?	Impairment?	Reversal?	
-----	-----	-----	-----	
Goodwill	No [OK] Yes	No		
Capitalized development	[OK] Yes	[OK] Yes	[OK] (IFRS only)	

6) How this affects financials (intuition)

- No amortization → profits look higher in normal periods
- But:
- Sudden large impairment → big earnings drop

-> This creates:

- "Cliff" effects in earnings
-

7) Clean mental model

- Goodwill = "premium paid for expectations"
 - Impairment = "those expectations didn't materialize"
-

If you want, I can show how companies like tech firms report massive goodwill impairments after bad acquisitions — those are some of the clearest real-world cases.

Turn 11: User

Time: 2026-05-01 16:53:00 GMT+8

has apple experienced goodwill impairments? report specific examples

Turn 12: Assistant

Time: 2026-05-01 16:53:03 GMT+8

Short answer: ****Apple has not reported meaningful goodwill impairment charges in recent years (and historically, essentially none)**** — which actually makes it a useful contrast vs companies that do impair goodwill.

Let's go through this clearly with evidence and context.

1) What Apple itself reports

From Apple's SEC filings:

- Apple tests goodwill annually for impairment
- But:

it "did not recognize any impairment charges related to goodwill" in multiple years ([SEC](https://www.sec.gov/Archives/edgar/data/320193/000032019317000070/R9.htm?utm_source=chatgpt.com))

Independent analyses of recent reports (2021–2022) also confirm:

- No evidence of goodwill impairment in those years ([ResearchGate](https://www.researchgate.net/publication/377467465_Financial_statement_analysis_and_revenue_forecast_for_Apple_Inc?utm_source=chatgpt.com))
-

2) Why Apple doesn't have goodwill impairments

This is actually more interesting than examples.

Reason 1 — Apple does relatively small acquisitions

- Apple typically buys:
- small AI startups
- niche tech companies
- Not huge, risky mega-mergers

-> So goodwill balances are:

- Smaller
- Less risky

Reason 2 — Strong post-acquisition performance

Goodwill impairment happens when:

“expected synergies don’t materialize”

Apple usually:

- Integrates acquisitions into its ecosystem
- Extracts value (chips, AI, software features)

-> So:

- The acquired businesses don’t underperform enough to trigger impairments

Reason 3 — Conservative accounting + strong margins

Apple generates:

- Massive cash flow
- High profitability

-> That keeps:

- “recoverable amount” comfortably above carrying value

→ no impairment needed

□□ 3) Important nuance: Apple DOES have goodwill

Apple does record goodwill from acquisitions (e.g., tech startups) ([Corporate Professionals](https://www.corporateprofessionals.com/articles/goodwill-valuation-understanding-the-intangible-power-behind-business-success/?utm_source=chatgpt.com))

But:

- It just hasn’t needed to write it down

4) Why your original sentence still matters

Even though Apple doesn’t impair goodwill, many companies do.

Contrast example (important for intuition):

- Microsoft
- Acquired Nokia
- Recorded ~\$7.6B goodwill impairment (huge write-off) ([TAG - The Algebra Group](https://thealgebragroup.com/what-is-goodwill-impairment?utm_source=chatgpt.com))

-> That's the kind of case where:

- Earnings crash (impairment)
- Then later "recover" (no impairment or other adjustments)

And models can misread that recovery

☐ 5) What WOULD happen if Apple had goodwill impairment

Hypothetically:

- Goodwill = 20B
- Impairment = 5B

Then:

- Income statement: -5B hit
 - Balance sheet: goodwill ↓ 5B
 - Cash: unchanged (non-cash)
-

6) Key takeaway

- Apple:
 - Has goodwill
 - Tests it annually
 - Rarely (basically never) impairs it
 - Other companies:
 - Often impair goodwill after bad acquisitions
-

☐ Final intuition

Goodwill impairment is usually a signal of:

"We overpaid or things didn't go as expected"

Apple's lack of impairments suggests:

"Their acquisition strategy is disciplined and execution is strong"

If you want, I can pull **a few tech companies that did impair goodwill and compare them to Apple side-by-side** — that's very insightful for understanding acquisition quality.

Turn 13: User

Time: 2026-05-01 16:56:33 GMT+8

does impairment help at all? is it tax benefit? anything else?

Turn 14: Assistant

Time: 2026-05-01 16:56:35 GMT+8

Short answer: impairment is mostly bad news—it means value was lost. But it can have a few secondary benefits (tax, cleaner future earnings, signaling). Let's break it down without sugarcoating.

1) What impairment actually does (mechanically)

When you record an impairment:

- Asset ↓
- Earnings ↓ (non-cash loss)
- Equity ↓

-> It's basically:

"We admit this asset isn't worth what we thought."

2) Does impairment help? (the honest answer)

☐ Economically: no

- You already lost value
 - Impairment just recognizes the loss
-

[OK] But there are side effects that can help

3) Tax benefit (sometimes, but not always)

Under many tax systems:

- Impairment can be tax-deductible

-> That means:

- Lower taxable income
 - Lower taxes paid
-

BUT big caveat:

- Goodwill impairments are often NOT tax-deductible (depends on jurisdiction)
- Many countries decouple:
- accounting rules vs tax rules

So:

- Some impairments → tax shield
 - Others → no tax benefit at all
-

4) “Resetting the base” (important insight)

After impairment:

- Asset value is lower
- Future depreciation/amortization is lower

-> Result:

- Future earnings look better
-

Example

Before impairment:

- Asset = 100
- Amortization = 20/year

After impairment to 40:

- Amortization = 8/year

-> Future profit ↑ (even if business didn't improve)

5) Earnings “big bath” strategy (very real behavior)

Sometimes management uses impairment to:

Take a big loss now to make future periods look better

- Huge loss this year
- Cleaner, higher earnings later

This is called:

- "big bath accounting"
-

6) Signaling / credibility (subtle benefit)

Recording impairment can signal:

- "We're being honest"
- "We're cleaning up the balance sheet"

Markets sometimes prefer:

- one large write-off

over

- years of small disappointments
-

7) Improves certain ratios (ironically)

After impairment:

- Assets ↓

So:

- ROA (Return on Assets) ↑ in future periods
- Asset turnover ↑

-> Makes the company look more efficient

8) Cash flow impact

Important:

Impairment is non-cash

- No cash leaves the company
 - Cash flow statement adds it back
-

9) IFRS vs US GAAP nuance

Under IFRS:

- Some impairments can be reversed later (except goodwill)

Under US GAAP:

- Generally no reversals

-> So:

- IFRS → more “bounce effects” in earnings
 - GAAP → more one-way downward
-

10) Clean mental model

Think of impairment as:

“Admitting past over-optimism”

Costs:

- Lower current earnings
- Lower equity

Possible side benefits:

- Tax savings (sometimes)
 - Higher future earnings (lower amortization)
 - Cleaner balance sheet
 - Better-looking ratios later
-

11) Bottom line

- It doesn't create value
- It just changes timing and presentation of losses

But because accounting affects perception:

It can still influence how investors (or models) interpret performance

If you want, I can show a real company case where impairment made future ROA and margins look artificially better — that's a really powerful intuition builder.

Turn 15: User

Time: 2026-05-01 17:00:16 GMT+8

for that bad nokia acquisition, why cant msft just keep carrying the goodwill? there is no real benefit of writing it off other than making the ROA look better. also what is asset turnover?

Turn 16: Assistant

Time: 2026-05-01 17:00:16 GMT+8

Good instinct to question it — but the key point is: ****they can't just keep carrying goodwill if it's no longer justified****. This isn't optional "cosmetic cleanup"; it's required by accounting rules.

1) Why Microsoft couldn't just keep the Nokia goodwill

After the Microsoft acquisition of Nokia Devices and Services, Microsoft had a large amount of goodwill on its balance sheet.

Under both:

- IFRS
- US GAAP

companies must:

Test goodwill for impairment at least annually

What the test asks:

"Is this asset still worth what we say it is?"

If the answer is no, then:

-> You are required to write it down

What happened with Nokia

- Microsoft expected strong smartphone performance
- The business underperformed badly
- Future cash flow expectations dropped

-> Result:

- ~\$7.6B goodwill impairment (massive loss)

☐ Why they can't just ignore it

If they didn't impair:

- Assets would be overstated
- Earnings would be misleading
- Financial statements would violate accounting standards

-> Auditors would not sign off -> Investors would be misled

So it's not:

"Should we take the loss?"

It's:

"We already lost it — now we must recognize it"

2) Is there any benefit to writing it off?

You're right that there's no economic gain — but a few indirect effects:

✓ Reality check (most important)

- Forces recognition of a bad investment

✓ Credibility

- Markets often prefer honesty over denial

✓ Future optics

- Lower asset base → ratios like ROA improve

But again:

These are side effects — not the reason it happens

3) Your intuition (very important)

You said:

"there is no real benefit other than making ROA look better"

That's actually mostly correct.

But flip the perspective:

It's not done to improve ROA

It's done because the asset is no longer real

4) What is asset turnover?

Asset turnover measures:

How efficiently a company uses assets to generate revenue

Formula

$$\text{Asset Turnover} = \frac{\text{Revenue}}{\text{Total Assets}}$$

5) Intuition

- High asset turnover → very efficient
- Low asset turnover → asset-heavy / less efficient

Example

- Revenue = 200
- Assets = 100

→ Asset turnover = 2.0

-> Means:

- Company generates \$2 of revenue per \$1 of assets

6) Why impairment affects asset turnover

After impairment:

- Assets ↓
- Revenue = unchanged

-> Asset turnover ↑

Example

Before impairment:

- Revenue = 200
- Assets = 100 → turnover = 2.0

After impairment:

- Assets = 70 → turnover = 2.86

-> Looks like:

“Wow, efficiency improved!”

But actually:

“We just admitted some assets weren’t worth what we thought”

7) Final mental model (this ties everything together)

- Goodwill = expectations about future performance
- Impairment = those expectations failed

And metrics like:

- ROA
- Asset turnover

can improve after impairment...

even though the business actually got worse

If you want, I can show a full before/after financial statement (income + balance sheet + ratios) for the Nokia case — that’s where this becomes extremely clear.

Turn 17: User

Time: 2026-05-01 17:03:27 GMT+8

sure

Turn 18: Assistant

Time: 2026-05-01 17:03:28 GMT+8

Great—let’s walk through a clean, simplified before/after of the Nokia situation so you can see how the accounting flows through everything.

Step 1) Before impairment (what Microsoft expected)

After the Microsoft acquisition of Nokia Devices and Services:

Balance Sheet (simplified)

- Revenue: 100
 - Other assets: 80
 - Goodwill: 40
 - -> Total assets: 120
-

Income Statement

- Revenue: 100
 - Expenses: 80
 - -> Net income: 20
-

Key ratios

ROA:

$$\sqrt[2]{20 / 120 = 16.7\%}$$

Asset turnover:

$$\sqrt[2]{100 / 120 = 0.83}$$

☐ Step 2) Impairment happens (reality hits)

Smartphone business underperforms → goodwill is overstated

-> Microsoft records impairment:

- Goodwill: 40 → 10
 - Impairment loss: 30
-

☐ Step 3) During impairment year

Income Statement

- Revenue: 100
- Operating expenses: 80

- Impairment loss: 30
 - -> Net income: -10
-

Balance Sheet

- Other assets: 80
 - Goodwill: 10
 - -> Total assets: 90
-

Ratios (this looks ugly)

ROA:

$$\backslash [-10 / 90 = -11.1\% \backslash]$$

Asset turnover:

$$\backslash [100 / 90 = 1.11 \backslash]$$

-> Notice:

- Profit collapses (real signal)
 - Asset turnover improves (misleading signal)
-

☐ Step 4) After impairment (next year)

Assume business is still mediocre:

Income Statement

- Revenue: 100
 - Expenses: 80
 - -> Net income: 20 (back to "normal")
-

Balance Sheet

- Assets remain: 90 (lower base)
-

Ratios now

ROA:

$$\lfloor 20 / 90 = 22.2\% \rfloor$$

Asset turnover:

$$\lfloor 100 / 90 = 1.11 \rfloor$$

Step 5) What just happened (this is the key insight)

Compare:

Metric	Before	After
Net income	20	20
Assets	120	90
ROA	16.7%	**22.2% ↑**
Asset turnover	0.83	**1.11 ↑**

-> It looks like:

“The company became more efficient!”

But reality:

“We just admitted some assets were never worth that much”

□□ Step 6) Why models get fooled

A model might see:

- Year 1: big loss
- Year 2: profit rebound
- Ratios improve

-> Conclude:

“Strong recovery and efficiency improvement”

But actually:

- No real operational improvement
- Just accounting reset of asset base

□ Final mental model

- Impairment = shrinking denominator (assets)
- Ratios improve because:

- Same numerator (profit or revenue)
- Smaller denominator

☐ One-liner takeaway

****Impairment makes future performance look better by lowering the baseline—not by improving the business.****

If you want, next step I'd recommend is: -> how to adjust financials to undo this distortion (what analysts actually do)

Turn 19: User

Time: 2026-05-01 17:05:02 GMT+8

yes how do analysts adjust for this effect

Turn 20: Assistant

Time: 2026-05-01 17:05:03 GMT+8

Great—this is exactly where “reading financials like an analyst” starts to matter.

The goal is simple:

Separate real operating performance from accounting noise (like impairments).

Here are the main ways analysts adjust for impairment effects.

1) Remove impairment from earnings (normalize profit)

Impairments are:

- Non-cash
- One-time (usually)
- Not part of ongoing operations

-> So analysts often compute:

“Adjusted earnings”

- Add back impairment losses
 - Ignore reversal gains
-

Example

Reported net income:

- -10 (includes -30 impairment)

Adjusted:

- $-10 + 30 = 20$

-> Now you're looking at:

"What would profit be without accounting noise?"

2) Fix the denominator problem (rebuild assets)

This is the more subtle but critical step.

After impairment:

- Assets are artificially lower
- ROA and asset turnover look better

-> Analysts may:

Option A — Use average assets (multi-year)

- Smooths out the drop

Option B — Use pre-impairment asset base

- Keeps comparability across time

Intuition

Instead of:

"Assets suddenly became smaller"

Treat it as:

"We corrected a past overstatement"

3) Focus on cash flow instead of earnings

Impairment:

- Hits earnings
- Does NOT affect cash

-> Analysts often rely on:

- Operating cash flow
- Free cash flow

These are:

Much harder to distort with accounting adjustments

4) Use EBITDA (with caution)

EBITDA removes:

- Depreciation
- Amortization

But:

-> Impairment is usually NOT included in EBITDA

So:

- EBITDA already ignores impairment in many cases

But beware:

- EBITDA also ignores real costs
- So it's not a perfect fix

5) Track “return on invested capital” (ROIC) carefully

ROIC is more robust than ROA—but still affected.

Analysts may:

- Use invested capital before impairment
- Or adjust capital to reflect:
 - acquisition cost
 - not just current book value

6) Look at segment-level performance

Especially relevant for cases like the Nokia deal:

- Revenue trends
- Unit economics

- Market share

-> These tell you:

Whether the business actually improved

—not just the accounting

7) Explicitly model impairments as “one-off events”

In forecasting:

- Treat impairment as:
 - non-recurring
 - not part of future earnings
-

8) Reverse IFRS-specific distortions (advanced)

Under IFRS:

- Impairments (non-goodwill) can be reversed

Analysts often:

- Remove both impairments and reversals
 - Build a “clean earnings series”
-

9) Compare across companies consistently

Especially important when comparing:

- IFRS vs US GAAP

Adjust for:

- Capitalized development
- Impairments
- Reversals

-> So you're comparing:

“Economics” instead of “accounting choices”

10) The core playbook (what good analysts actually do)

Step 1:

- Strip out impairments from earnings

Step 2:

- Normalize asset base (avoid artificially low denominator)

Step 3:

- Cross-check with cash flow

Step 4:

- Focus on operating metrics (revenue, margins)
-

Final mental model

Think of impairments as:

Accounting shocks that distort both the numerator and denominator

So analysts:

- Fix the numerator → adjusted earnings
 - Fix the denominator → normalized assets
-

□ One-line takeaway

Good analysts don't take reported ROA or earnings at face value—they rebuild the numbers to reflect underlying economics.

If you want, I can show a step-by-step “adjusted ROA” calculation using the Nokia example—that's exactly how this is done in practice.

Turn 21: User

Time: 2026-05-01 17:51:52 GMT+8

can you explain operating cash flow and free cash flow? is free cash flow the inbound cash left after paying for capex and interests and operation?

Turn 22: Assistant

Time: 2026-05-01 17:51:53 GMT+8

You're on the right track—just need to tighten the definitions a bit. The key is separating operations vs investing vs financing.

1) Operating Cash Flow (OCF)

OCF = cash generated by the core business

-> Think:

“How much real cash does the business produce from selling its product/service?”

What's included

- Cash received from customers
 - Cash paid to suppliers & employees
 - Working capital changes (inventory, receivables, payables)
-

What's NOT included

- ☐ Capex (buying equipment, building software, etc.)
 - ☐ Interest payments (depends on accounting framework, but often excluded in analysis)
 - ☐ Financing (debt, equity)
-

Simple formula (intuition version)

$$\text{OCF} \approx \text{Net Income} + \text{Non-cash items} - \text{Working capital increases}$$

Key idea

OCF strips out accounting noise (like depreciation, impairments) and focuses on cash reality

2) Free Cash Flow (FCF)

Now we go one step further:

FCF = cash left after maintaining/expanding the asset base

Standard definition

$$\text{FCF} = \text{Operating Cash Flow} - \text{Capital Expenditures}$$

Intuition

- OCF = “cash generated”
- Capex = “cash reinvested to keep/grow the business”

-> FCF =

“Cash you can actually take out of the business”

3) Your question (important correction)

You asked:

“is free cash flow the inbound cash left after paying for capex and interests and operation?”

Almost correct, but:

☐ Standard FCF does NOT subtract interest

- Interest is usually treated as:
 - financing activity
 - not operating (in analysis)
-

So:

Typical FCF:

- [OK] subtract:
 - operating costs (already in OCF)
 - capex
 - ☐ do NOT subtract:
 - interest
 - debt repayments
-

4) Why interest is excluded

Because analysts want:

A measure independent of capital structure

Two identical businesses:

- One financed with debt
- One with equity

-> Should have same FCF (before financing decisions)

5) Variants (this is where confusion comes from)

There are two common versions:

A) Free Cash Flow to Firm (FCFF)

- Before interest
- Available to:
 - debt holders
 - equity holders

-> Most common in valuation

B) Free Cash Flow to Equity (FCFE)

- After interest and debt payments

-> This is closer to what you described

6) Example (ties it all together)

- OCF = 100
 - Capex = 30
 - Interest = 10
-

FCFF (standard FCF):

- $100 - 30 = 70$
-

FCFE:

- $100 - 30 - 10 = 60$
-

7) Why this matters (link to your earlier questions)

Remember impairments?

- They:
- crush net income
- do NOT affect cash

-> So:

- OCF → mostly unaffected
- FCF → mostly unaffected

That's why analysts trust:

Cash flow > earnings

8) Clean mental model

- OCF → "Is the business generating cash?"
- FCF → "After reinvesting, how much cash is left?"
- FCFE → "How much cash can equity holders actually take?"

□ One-line takeaway

Free cash flow is operating cash minus reinvestment (capex), not financing costs like interest—unless you're specifically looking at equity cash flow.

If you want, I can show how a company can have high earnings but negative free cash flow (super common in tech and startups).

Turn 23: User

Time: 2026-05-01 17:59:35 GMT+8

is capex considered investment cash flow? what about investing into short term securities? and do you get net cash balance after doing $ocf + fcf + icf$?

Turn 24: Assistant

Time: 2026-05-01 17:59:35 GMT+8

Good set of questions—this is exactly how the cash flow statement fits together.

1) Is capex considered investing cash flow?

Yes.

Capex (buying equipment, servers, buildings, capitalized software, etc.) is part of:

Cash Flow from Investing Activities (CFI)

Example

- Buy servers for \$50

→ CFI = -50

Even though we subtract capex when computing FCF:

$$[\text{FCF} = \text{OCF} - \text{Capex}]$$

-> In the actual cash flow statement:

- Capex lives in investing, not operating
-

2) What about short-term securities?

Also investing cash flow.

Example

- Buy Treasury bills or money market funds:
- Cash ↓
- Short-term investments ↑

→ CFI = negative

- Sell them later:
- Cash ↑

→ CFI = positive

Subtle nuance

Short-term securities are often called:

- “cash equivalents” or “marketable securities”

-> But in the cash flow statement:

- Buying/selling them still shows up in investing
-

3) Do you get net cash from $OCF + FCF + ICF$?

☐ Not quite—this is where the structure matters.

The correct equation

$$\Delta \text{Cash} = \text{OCF} + \text{CFI} + \text{CFF}$$

Where:

- OCF = operating cash flow
 - CFI = investing cash flow
 - CFF = financing cash flow
-

☐ Important:

- FCF is NOT a separate category in the cash flow statement
 - It's a derived metric, not an official line item
-

4) Why adding FCF would double-count

Since:

$$FCF = OCF - \text{Capex}$$

And capex is already inside CFI, if you do:

$$OCF + FCF + CFI$$

-> You're counting capex twice

5) Full example (ties everything together)

Let's say:

- $OCF = +100$
- $\text{Capex} = -30$ (in CFI)

- Buy securities = -20 (in CFI)
 - Debt raised = +10 (in CFF)
-

Cash flow statement

- OCF = +100
 - CFI = -50
 - CFF = +10
-

Net change in cash

$$\backslash[100 - 50 + 10 = +60 \backslash]$$

FCF (just for analysis)

$$\backslash[\text{FCF} = 100 - 30 = 70 \backslash]$$

-> Notice:

- FCF ignores the -20 securities purchase
 - Because that's just reallocating cash, not "using it up"
-

6) Key intuition

- OCF → cash generated by business
 - CFI → where cash is invested (capex, securities, acquisitions)
 - CFF → how business is funded (debt, equity, dividends)
-

7) Clean mental model

Think of cash flow like:

- OCF = "money coming in from operations"
- CFI = "what we did with it (investments)"
- CFF = "how we financed everything"

And:

FCF = a shortcut metric, not part of the official flow

□ Final takeaway

- [OK] Capex → investing cash flow
- [OK] Short-term securities → investing cash flow
- □ You don't sum OCF + FCF + CFI
- [OK] You sum:

OCF + CFI + CFF = change in cash

If you want, I can map this to a real Apple or Microsoft cash flow statement line-by-line—that makes it very concrete.

Turn 25: User

Time: 2026-05-01 18:03:56 GMT+8

in schilit's financial shenanigans, he mentioned one shenanigan related to inflating ocf by taking on debt, is that correct? how would it work?

Turn 26: Assistant

Time: 2026-05-01 18:03:57 GMT+8

Yes—Schilit flags this as a classic classification trick: making operating cash flow (OCF) look stronger by pulling in cash that really belongs to financing.

Under normal rules (both IFRS and US GAAP):

- Borrowing cash → Financing cash flow (CFF)
- OCF should reflect only operating activities

So if OCF “jumps” because of something that smells like borrowing, that’s a red flag.

How companies can (misleadingly) boost OCF with “debt”

1) Supplier financing / reverse factoring (big one)

What it is

- The company arranges for a bank to pay its suppliers.
- The company now owes the bank later (effectively short-term debt).

What changes in the cash flow statement

- Payments to suppliers are delayed, so:

- Accounts payable ↑
- OCF ↑ (less cash going out today)

Why this is like debt

- Economically, the company borrowed from the bank to pay suppliers.

Why it inflates OCF

- The increase in payables shows up as a source of operating cash, even though the real driver is financing.

-> Schilit's point: it looks like stronger operations, but it's really hidden borrowing.

2) Aggressively stretching payables

What it is

- Simply paying suppliers later (without a formal program)

Effect

- Accounts payable ↑ → OCF ↑

Why it's similar

- You're effectively using suppliers as a financing source
-

3) Selling receivables (factoring)

What it is

- Sell accounts receivable to a bank/factor to get cash early

Two treatments

- Proper: shown as financing or investing
- Aggressive: classified as operating inflow

Effect

- Cash comes in → OCF ↑

Reality

- It's often closer to borrowing against receivables
-

4) Customer advances (sometimes legitimate, sometimes abused)

What it is

- Collect cash upfront from customers

Effect

- Deferred revenue $\uparrow \rightarrow$ OCF $\uparrow$

When it's a problem

- If driven by heavy discounting or one-off deals just to pull cash forward
-

Why this matters

All of these create the illusion:

"The business is generating strong cash!"

But reality:

"The company is just delaying payments or borrowing indirectly"

Quick diagnostic checklist (what analysts look for)

☐ Red flags

- OCF growing faster than revenue
 - Big increases in:
 - Accounts payable
 - "Other liabilities"
 - Mentions of:
 - "supplier financing"
 - "structured payables"
 - OCF high, but debt also rising
-

Clean example

Normal case

- OCF = 100
 - Pay suppliers normally
-

With supplier financing

- Bank pays suppliers
- Company delays payment

Result:

- OCF = 130 (looks great)

- Hidden debt ↑

-> No real improvement in operations

Key takeaway

OCF can be inflated not just by better business—but by shifting financing activities into operating-looking flows.

That's exactly the kind of thing Schilit calls a "financial shenanigan."

If you want, I can show a real company example (like how this showed up in filings)—those are very eye-opening.

Turn 27: User

Time: 2026-05-01 18:34:43 GMT+8

Example 1: US market rules for foreign private issuers The SEC says:

- domestic issuers use U S GAAP
- foreign private issuers may use IFRS as issued by the IASB
- or home-country GAAP with reconciliation to US GAAP

Why this matters:

- a company listed in the US is not necessarily on US GAAP
- a foreign company may appear in the same market data universe but use a different

accounting basis ML trap: If I group companies by exchange alone, I may accidentally mix domestic US GAAP issuers with IFRS filers in the same feature space.

who are examples of foreign private issuers? and how can private issuers trade publicly?

Turn 28: Assistant

Time: 2026-05-01 18:34:43 GMT+8

Great question—this gets into how global companies access U.S. markets without becoming "U.S. companies" for reporting purposes.

1) What is a "foreign private issuer" (FPI)?

A foreign private issuer (FPI) is:

A company incorporated outside the U.S. that meets SEC criteria allowing it to follow a different reporting regime than U.S. domestic issuers.

Under the U.S. Securities and Exchange Commission rules:

- FPIs can use:
- IFRS (as issued by IASB), or
- their home-country GAAP (with reconciliation, historically)

-> They don't have to use US GAAP

2) Examples of foreign private issuers

These are large, well-known companies trading in U.S. markets but not using US GAAP:

China Alibaba Group

- Listed on NYSE (ADR)
 - Reports under IFRS
 - Clearly a foreign private issuer
-

□□ SAP

- Listed in the U.S.
 - Uses IFRS
-

□□ BP

- Trades in the U.S. (ADR)
 - Uses IFRS
-

□□ Nestlé

- U.S.-traded ADRs
 - Uses IFRS
-

□□ SK Telecom

- U.S. listing via ADR
 - IFRS-based reporting (K-IFRS)
-

-> Notice:

These all trade in the same U.S. market data universe as U.S. companies—but follow different accounting rules

3) How can “private issuers” trade publicly?

The term is confusing.

“Private” here does NOT mean “not publicly traded”

It means:

Not a U.S. domestic issuer

Two main ways they trade in the U.S.

1) ADRs (most common)

American Depositary Receipts

- A U.S. bank holds the foreign company’s shares
- Issues receipts that trade on U.S. exchanges

-> Example:

- Alibaba ADR on NYSE
- Each ADR = claim on underlying shares in China

2) Direct listing on U.S. exchanges

Some foreign companies:

- List shares directly on NYSE/Nasdaq
- Still qualify as FPIs

4) Why FPIs get special treatment

The SEC allows flexibility because:

- Forcing US GAAP would:
- discourage foreign listings
- increase compliance cost

So FPIs get:

- IFRS allowed
 - Less frequent reporting (e.g., annual Form 20-F instead of 10-K/10-Q)
-

5) Why this matters for your ML point (very important)

You wrote:

"If I group companies by exchange alone..."

That's exactly the trap.

What goes wrong

You might mix:

Company	Accounting
-----	-----
Apple	US GAAP
Alibaba	IFRS
SAP	IFRS

But features like:

- Earnings
- Assets
- R&D
- Impairments

-> are not directly comparable

Example distortion

- IFRS company capitalizes development → higher assets
- US GAAP company expenses → lower assets

-> Your model might learn:

"Company A is more efficient"

When really:

It's just accounting differences

6) Clean mental model

- Domestic issuer = U.S. company → US GAAP
- Foreign private issuer = non-U.S. company → IFRS (often)

And:

“Private” = foreign status, not ownership structure

□ Final takeaway

A stock trading in the U.S. does NOT imply US GAAP—foreign private issuers often use IFRS, which can materially distort cross-company comparisons (and ML models).

If you want, I can show how to automatically detect IFRS vs GAAP companies in a dataset—that’s super useful for building clean ML pipelines.

Turn 29: User

Time: 2026-05-01 18:41:13 GMT+8

yes please, and how can i account for these differences for my ml model? do i use adjusted earnings? and what are other differences in the acct standards between us gaap and ifrs that i should be aware of?

Turn 30: Assistant

Time: 2026-05-01 18:41:14 GMT+8

Great—this is exactly the right way to think about it: don’t let accounting regimes leak into your model as “signal.” You want to either normalize or explicitly model those differences.

1) How to detect IFRS vs US GAAP in data

A. Use filing metadata (most reliable)

- U.S. domestic issuers → Form 10-K / 10-Q → US GAAP
- Foreign private issuers → Form 20-F / 6-K → often IFRS

B. Look for explicit disclosures

In filings/notes:

- “Prepared in accordance with IFRS as issued by the IASB”
- “Prepared in accordance with U.S. GAAP”

C. Programmatic flags (practical)

- Exchange ≠ enough
- Use:
- filing type
- country of incorporation
- “accounting standard” field (if your data vendor has it)

-> In your feature table, create:

- `accounting_standard` {US_GAAP, IFRS}
-

2) Strategy for ML: two robust approaches

Approach 1 — Make accounting regime a feature

- Add a binary/categorical feature:
- `is_ifrs = 1/0`

-> Pros:

- Simple
- Lets model learn regime differences

-> Cons:

- Model may overfit to accounting artifacts
-

Approach 2 — Normalize key metrics (recommended)

You rebuild features so IFRS and GAAP become comparable.

This is what serious analysts do.

3) Core normalization adjustments (high impact)

A. R&D / Development costs (biggest one)

- IFRS → can capitalize development
- GAAP → mostly expense all R&D

Adjustment:

- Treat IFRS development like GAAP:
- Expense capitalized development

- Remove it from assets

-> Effect:

- Lower assets
 - Lower earnings (initially)
 - Better comparability
-

B. Impairments & reversals

- IFRS → allows reversals (except goodwill)
- GAAP → no reversals

Adjustment:

- Remove:
- impairment losses
- impairment reversals

-> Use:

- “core earnings” = earnings excluding impairments
-

C. Lease accounting (less of an issue now, but still relevant)

- Both standards now capitalize leases, but:
- classification and presentation still differ slightly

Adjustment:

- Standardize:
 - lease liabilities
 - lease-related expenses
-

D. Interest classification (cash flow)

Under IFRS:

- Interest paid can be:
- operating or financing

Under US GAAP:

- Interest paid → operating

Adjustment:

- Reclassify to a consistent definition:
 - move all interest to financing (common in analysis)
-

E. Inventory (LIFO vs FIFO)

- GAAP → allows LIFO
- IFRS → prohibits LIFO

Adjustment:

- If possible:
 - convert LIFO → FIFO using LIFO reserve
-

F. Goodwill & intangibles

- Both:
- no amortization of goodwill
- impairment-only

But:

- IFRS may allow more capitalization of intangibles

Adjustment:

- Be cautious comparing:
 - asset levels
 - ROA
-

4) Feature design that works well in ML

Instead of raw accounting numbers, use:

More robust features

- Cash-based:
- Operating cash flow
- Free cash flow
- Margins:
- Operating margin (adjusted)
- Growth rates:

- Revenue growth
 - Cash flow growth
-

Avoid raw:

- Net income (unadjusted)
 - Total assets (without normalization)
-

5) Example: adjusted earnings feature

Yes—you asked:

“do i use adjusted earnings?”

-> Yes, but carefully defined

Example:

```
adjusted_net_income =  
    net_income  
    + impairment_losses  
    - impairment_reversals  
    - capitalized_development
```

6) Best-practice ML pipeline (practical)

Step 1 — Tag regime

- is_ifrs

Step 2 — Normalize major distortions

- R&D capitalization
- Impairments
- Interest classification

Step 3 — Build robust features

- Cash flow-based
- Margin-based
- Growth-based

Step 4 — Sanity checks

- Compare distributions:
- IFRS vs GAAP
- They should look similar after adjustment

7) Other important IFRS vs GAAP differences (quick list)

These can matter depending on your use case:

Revenue recognition

- Mostly aligned now, but small differences remain

Provisions (liabilities)

- IFRS → more flexible (principles-based)
- GAAP → stricter thresholds

-> IFRS may show:

- more liabilities earlier

Asset revaluation

- IFRS → allows revaluation of some assets
- GAAP → generally historical cost only

-> IFRS assets can be:

- higher (not purely operational)

Development costs

- Already covered, but very important

8) Clean mental model

You're trying to move from:

“Reported accounting numbers”

to:

“Underlying economic reality”

☐ Final takeaway

For ML:

Don't rely on raw accounting metrics across IFRS and GAAP—either normalize key differences (R&D, impairments, cash flow classification) or explicitly model the accounting regime.

If you want, I can help you design a feature schema tailored to Bubblic-style datasets (e.g., ranking companies, forecasting, anomaly detection)—that's where this becomes very practical.

Turn 31: User

Time: 2026-05-01 18:55:05 GMT+8

for the r&d example, how would the capitalized r&d show up on the balance sheet? so is it my job to find this element when it first appears and expense it that year? and subsequent years would just remove the remaining from asset?

Turn 32: Assistant

Time: 2026-05-01 18:55:05 GMT+8

You're thinking about it exactly the right way. Let's make it concrete and precise.

1) Where capitalized R&D shows up on the balance sheet

Under IFRS (via IAS 38):

Capitalized development costs appear as:

Intangible assets

Typical labels you'll see:

- "Capitalized development costs"
- "Internally generated intangible assets"
- Sometimes buried inside:
- "Intangible assets, net"

Example (balance sheet)

Assets Amount
----- -----
Cash 50
PPE 100
Intangible assets (incl. dev) 80

Inside that 80:

- say 30 = capitalized development
-

2) How it flows over time (very important)

Year 1 (development spending)

- Spend 100 on development
- Capitalize it

-> Balance sheet:

- Intangible asset = +100

-> Income statement:

- No expense (or minimal)
-

Year 2+ (amortization phase)

- Amortize, say 20/year

-> Each year:

- Expense = 20
 - Asset decreases: 100 → 80 → 60 → ...
-

3) What you're trying to do (for ML normalization)

You said:

"is it my job to find this element when it first appears and expense it that year?"

-> Yes — that's the core idea

You want to convert IFRS → "GAAP-like" treatment:

Expense development immediately instead of capitalizing

4) Practical ways to implement this

There are two levels of sophistication

☐ Method A (simpler, very practical)

Use cash flow statement:

Look for:

- “Capitalized development costs” (investing outflow)

-> Then:

$$\text{adjusted_expense} = \text{reported_R\&D} + \text{capitalized_development}$$

-> This:

- Converts IFRS behavior → GAAP-like
-

☐ Method B (more precise, harder)

Rebuild the asset roll-forward:

$$\text{Ending Dev Asset} = \text{Beginning} + \text{Capitalized}$$

- Amortization
- Impairment

$$\text{}$$

Then:

- Identify new capitalization each year
 - Expense it immediately
 - Remove amortization (since you already expensed upfront)
-

5) Your second question (important nuance)

“subsequent years would just remove the remaining from asset?”

-> Yes—but carefully:

If you expense everything upfront:

Then you should also:

- ☐ Remove amortization expense (since it's double counting)
 - ☐ Remove the asset from the balance sheet
-

Intuition

You're replacing:

“spread cost over time”

with:

“recognize all cost immediately”

6) Clean transformation logic

Income statement

Adjusted earnings =
Reported earnings
+ amortization of dev costs
– capitalized development (current year)

Balance sheet

Adjusted assets =
Reported assets
– capitalized development asset

7) Why this works

You’re making IFRS behave like US GAAP:

- No capitalization
- No future amortization
- All cost recognized upfront

8) Practical shortcut (what many quants actually do)

If full reconstruction is hard:

- Use:
- cash flow “capitalized development” line
- OR “additions to intangibles”

Then:

- Treat it as:
- operating expense (not investing)

9) Key pitfalls

☐ ☐ 1. Not all intangibles = R&D

- Could include:
- acquired IP
- licenses

-> Try to isolate:

- “internally generated” or “development”

☐ ☐ 2. Missing disclosures

- Some companies don't break it out cleanly

-> You may need:

- footnotes parsing (important for ML pipeline quality)

10) Final mental model

You're converting:

IFRS view	Your normalized view
-----	-----
"This creates future value"	"Treat as current expense"
Asset grows	No asset
Future amortization	No amortization

☐ One-line takeaway

Yes—identify capitalized development each year, expense it immediately, and remove the associated asset and amortization to make IFRS comparable to GAAP.

If you want, I can show a real company filing snippet and how to extract this programmatically (e.g., from 20-F footnotes)—that’s where this becomes very actionable.

Turn 33: User

Time: 2026-05-01 19:01:51 GMT+8

the capital development amount appears as investment outflow in cash flow statement for both ifrs and gaap?

also, why would quants put that under operating cf not investing cf?

Turn 34: Assistant

Time: 2026-05-01 19:01:52 GMT+8

Great questions—this is where accounting classification vs economic reality diverge.

1) Does capitalized development show up as investing outflow in both IFRS and GAAP?

Under IFRS:

- Yes — capitalized development costs are typically shown as:

Investing cash outflow (because they create an asset)

Under US GAAP:

- Usually not applicable, because:
- Most R&D is expensed immediately
- So it shows up in:

Operating cash flow (OCF)

□ So the asymmetry is:

Treatment Cash flow classification
----- -----
IFRS (capitalized dev) Investing (CFI)
GAAP (expensed R&D) Operating (OCF)

-> Same economic activity → different sections of cash flow statement

2) Why this creates a problem

Two identical companies:

- Company A (IFRS):
 - Capitalizes development
 - OCF looks higher
 - CFI more negative
- Company B (GAAP):
 - Expenses R&D
 - OCF looks lower

-> A naive model concludes:

“Company A generates more operating cash”

But that's just accounting classification

3) Why quants move it into operating cash flow

This is the key idea:

R&D is part of core operations, not investing in the economic sense

Think about what “investing” usually means

- Buying factories
- Acquiring companies
- Buying securities

These are:

Optional, discrete capital allocation decisions

But R&D / development is different

- Continuous
- Necessary to stay competitive
- Core to the business (especially tech/pharma)

-> Economically:

It behaves like an operating expense, not optional investment

4) What quants actually do

They “fix” the inconsistency by reclassifying:

Adjustment

$$\text{Adjusted OCF} = \text{Reported OCF} - \text{Capitalized Development}$$

$$\text{Adjusted CFI} = \text{Reported CFI} + \text{Capitalized Development}$$

Effect

- IFRS company now looks like GAAP company
 - Apples-to-apples comparison
-

5) Intuition (very important)

Accounting says:

“We created an asset → investing activity”

Quants say:

“You paid engineers to build product → that’s operations”

6) Why this matters for ML (your use case)

If you don’t adjust:

Your model learns:

- IFRS → high OCF
- GAAP → low OCF

-> It will incorrectly infer:

“IFRS companies are better at generating cash”

7) Clean mental model

- Accounting classification = arbitrary (to some extent)
 - Economic function = what you care about
-

8) One-line takeaway

Capitalized development is classified as investing under IFRS, but quants often move it to operating cash flow because economically it behaves like a recurring operating expense.

If you want, next step I’d suggest is: -> building a fully normalized cash flow (OCF/CFI/CFF) schema for ML—this is where your pipeline becomes very robust.

Turn 35: User

Time: 2026-05-01 19:04:12 GMT+8

ah i see, the investment cf would be consistent between those standards for capex related to PPE right?

Turn 36: Assistant

Time: 2026-05-01 19:04:12 GMT+8

Yes—mostly yes, but with a couple of nuances worth knowing.

1) Core idea: PPE capex is consistent

For spending on physical assets (factories, servers, equipment):

- Under IFRS
- Under US GAAP

-> Capex for PPE shows up as investing cash outflow (CFI)

Example

- Buy servers for \$50

Cash flow statement:

- CFI = -50

-> This is consistent across both standards

2) Why this one is consistent

Unlike R&D:

- PPE is clearly:
- Long-lived
- Tangible
- Capital investment

-> No ambiguity → both standards treat it the same

3) Subtle differences you should still be aware of

Even though classification is consistent, measurement and timing can differ

A) Capitalization thresholds / policies

Companies may differ in:

- What qualifies as capex vs expense
- Internal policies (e.g., minimum dollar thresholds)

-> Effect:

- One company capitalizes more → higher assets, higher CFI
-

B) Timing differences

- When is an asset recognized?
- Delivery date?
- When ready for use?

-> Can shift capex between periods

C) Leases (important edge case)

Both standards now capitalize leases, but:

- Lease-related cash flows may appear partly in:
 - operating
 - financing

-> So:

- “Capex-like” activity (right-of-use assets) doesn’t always sit cleanly in CFI
-

D) Asset revaluation (IFRS only)

Under IFRS:

- Companies can revalue PPE upward

Under US GAAP:

- No revaluation (historical cost)

-> This affects:

- Asset base
 - ROA
- but not the cash flow classification

4) Big picture comparison

Item	IFRS	US GAAP	Comparable?	
PPE capex	Investing	Investing	[OK] Yes	
R&D	Investing (if capitalized)	Operating	No	
Interest	Flexible	Operating	No	

5) What this means for your ML pipeline

Good news:

PPE capex is relatively “safe” to compare across companies

So:

- You can use:
 - Capex
 - Capex / revenue
 - Capex / assets
- > With less normalization needed than R&D

6) Clean mental model

- PPE capex → universally considered “investment”
- R&D → accounting disagrees on whether it’s investment or expense

□ Final takeaway

Yes—PPE-related capex is consistently classified as investing cash flow under both IFRS and US GAAP, making it far more comparable than R&D-related cash flows.

If you want, I can show a “safe vs unsafe features” list for ML using financial statements—that’s super useful when you start building models.

Turn 37: User

Time: 2026-05-01 19:07:42 GMT+8

does lease refer to company leasing its property out or company renting a building to occupy it?

Turn 38: Assistant

Time: 2026-05-01 19:07:42 GMT+8

“Lease” can mean both, depending on which side the company is on.

1) Two perspectives

A) Company renting something → lessee

The company is using an asset owned by someone else

Examples:

- Renting office space
- Leasing servers or equipment
- Leasing retail stores

-> This is what most people mean in practice

B) Company leasing something out → lessor

The company owns the asset and rents it to others

Examples:

- Real estate company renting buildings
 - Equipment leasing company
 - Car leasing business
-

2) What accounting usually focuses on

When you hear lease accounting in financial statements (especially for tech companies):

-> It's almost always about the lessee side (i.e., the company renting space or equipment)

3) How lessee accounting works today

Under both:

- IFRS
- US GAAP

-> Companies must put most leases on the balance sheet

What gets recorded

When a company rents a building:

- Right-of-use (ROU) asset → “we have the right to use this”
 - Lease liability → “we owe future rent payments”
-

Example

Company signs a 5-year office lease:

Balance sheet:

- Asset: ROU asset
 - Liability: lease obligation
-

4) Why this matters (link to your earlier questions)

Leases behave like:

“hidden debt + hidden asset”

So they affect:

- Assets (↑)
 - Liabilities (↑)
 - Cash flow classification (split across operating & financing)
-

5) Cash flow nuance

Lease payments are split:

- Part = interest → often operating
- Part = principal → financing

-> So:

- Not fully in investing (like capex)
- Not purely operating either

6) Clean mental model

- Lessee (most common) → “I’m renting → I record asset + liability”
- Lessor → “I’m the landlord → I earn rental income”

□ Final takeaway

In most financial analysis contexts, “lease” refers to a company renting assets (lessee), not leasing them out—though both exist.

If you want, I can show how leases can artificially boost or distort OCF and EBITDA—that’s another common modeling pitfall.

Turn 39: User

Time: 2026-05-01 19:10:02 GMT+8

for principal and interest payment on loans, those are all financing cf right?

Turn 40: Assistant

Time: 2026-05-01 19:10:03 GMT+8

Short answer: not exactly—principal is financing, interest depends on the accounting standard (and choice under IFRS).

1) Principal vs interest — the clean split

[OK] Principal repayment

- Always:

Financing cash flow (CFF)

-> Because it changes the amount of debt outstanding

☐☐ Interest payment

This is where it differs:

Under US GAAP

- Interest paid → Operating cash flow (OCF)
- Interest received → Operating cash flow

-> Fixed, no choice

Under IFRS

Companies have flexibility:

- Interest paid → Operating OR Financing
- Interest received → Operating OR Investing

-> Must be consistent over time

2) Why this distinction exists

- Principal = returning borrowed money → financing
 - Interest = cost of doing business (debatable):
 - GAAP → treat as operating expense
 - IFRS → allow judgment
-

3) Example

Company pays:

- Principal: 50
 - Interest: 10
-

Under US GAAP:

- OCF: -10 (interest)
 - CFF: -50 (principal)
-

Under IFRS (one possible choice):

- OCF: 0
 - CFF: -60 (principal + interest)
-

4) Why this matters (important for your ML work)

Two identical companies:

- Same operations
- Same debt

-> But:

Metric	GAAP	IFRS
OCF	lower	higher
CFF	lower	more negative

Result:

Your model might think:

"IFRS company generates more operating cash"

But:

It's just classification

5) What analysts (and you) should do

Normalize:

-> Move all interest to financing

So:

Adjusted OCF = Reported OCF + interest paid
Adjusted CFF = Reported CFF - interest paid

6) Clean mental model

- Principal → always financing
 - Interest → operating (GAAP) vs flexible (IFRS)
-

□ Final takeaway

Principal repayments are always financing cash flows, but interest payments are operating under US GAAP and flexible under IFRS—so you often need to normalize them for comparability.

If you want, I can give you a full “cash flow normalization checklist” for ML pipelines—that would tie together everything we’ve discussed.

Turn 41: User

Time: 2026-05-01 19:13:14 GMT+8

why not move the interest from fcf to ocf to normalize to us gaap?

Turn 42: Assistant

Time: 2026-05-01 19:13:16 GMT+8

Good question—and this gets to a subtle but important point: there isn’t one “correct” normalization. It depends on what you want your metric to represent.

1) Yes—you can move interest into OCF (GAAP-style)

If your goal is:

“Make everything look like US GAAP”

then you would:

Adjusted OCF = Reported OCF – interest_paid (if IFRS classified it in financing)
Adjusted CFF = Reported CFF + interest_paid

-> This aligns IFRS companies with GAAP reporting.

2) But many analysts do the opposite (and here’s why)

They move interest out of OCF into financing, even for GAAP companies.

Why?

They want a measure of operating performance independent of capital structure

Key idea

Interest depends on:

- how much debt you have
- your financing choices

NOT on:

- how good your core business is
-

So analysts prefer:

“OCF before financing decisions”

Which means:

- exclude interest from OCF
-

3) Two valid normalization philosophies

A) Accounting-consistent (GAAP-style)

- Interest → OCF
- Matches reported GAAP numbers

-> Good for:

- Reproducing reported metrics
 - Simpler pipelines
-

B) Economic-consistent (analyst-style)

- Interest → financing
- OCF reflects pure operations

-> Good for:

- Comparability across firms
 - ML models focused on business quality
-

4) Why your earlier FCF definition matters

Recall:

$$\text{FCF} = \text{OCF} - \text{Capex}$$

If OCF includes interest:

-> FCF becomes:

“cash available after debt costs”

But analysts often want:

“cash generated before financing decisions”

5) Example (shows the difference clearly)

Two identical companies:

- Same operations
- One uses debt, one doesn't

If interest is in OCF:

Company	OCF
----- -----	
No debt	100
With debt	90

-> Looks like:

worse operations

If interest is moved to financing:

Company	OCF
----- -----	
No debt	100
With debt	100

-> Now:

operations look identical (correct)

6) What quants usually do (practically)

For ML:

-> Most robust choice:

- Move interest to financing
- Use:
- OCF (pre-interest)
- FCF (pre-interest)

Because:

It removes capital structure noise

7) So why NOT normalize to GAAP?

Because GAAP itself:

mixes operating performance with financing decisions

Which is fine for reporting—but not ideal for modeling.

8) Clean mental model

- GAAP OCF = operations + financing cost (interest)
- Analyst OCF = operations only

□ Final takeaway

****You can move interest into OCF to match US GAAP, but most analysts do the opposite—moving interest out of OCF—because they want a cleaner measure of operating performance independent of financing decisions.****

If you want, I can show how this choice affects valuation models (DCF, multiples)—that's where it really becomes critical.

Turn 43: User

Time: 2026-05-01 19:16:36 GMT+8

ok

Turn 44: Assistant

Time: 2026-05-01 19:16:38 GMT+8

Great—let's connect this directly to valuation, because that's where the “move interest or not?” decision really matters.

1) Two valuation frameworks (this is the root of everything)

A) Firm valuation (most common)

You value the entire business:

Value = Enterprise Value (EV)

Use:

- Free Cash Flow to Firm (FCFF)
 - Discount rate = WACC
-

B) Equity valuation

You value just the equity:

Value = Market cap

Use:

- Free Cash Flow to Equity (FCFE)
 - Discount rate = cost of equity
-

2) Where interest fits

For FCFF (firm view)

-> Exclude interest

Why:

- You're valuing the business before financing decisions

So:

- Interest → treated as financing
 - Not in OCF
 - Not in FCF
-

For FCFE (equity view)

-> Include interest

Why:

- Equity holders only get cash after debt holders are paid

So:

- Interest → included
- Debt repayments → included

3) This explains your earlier question

You asked:

“Why not move interest into OCF (GAAP style)?”

Answer:

- If you do that → you’re implicitly moving toward equity-style cash flow
- If you remove it → you’re building firm-style cash flow

4) Concrete example (very important)

Two identical companies:

- Same operations
- One has debt, one doesn’t

Case 1 — Using FCFF (no interest)

Metric	Both companies
OCF (adjusted)	100
Capex	30
FCF	70

-> Correct:

Same business → same value

Case 2 — Using GAAP-style OCF (includes interest)

Metric	No debt	With debt
OCF	100	90
FCF	70	60

-> Looks like:

leveraged company is worse

But:

it's just financing structure

5) Why ML models break here

If you don't normalize:

Your model learns:

- "Companies with less debt = better OCF"

But that's not:

- operational performance

It's:

- capital structure
-

6) Best practice for ML (very actionable)

-> Use firm-style features

- OCF before interest
 - FCF before interest
 - Separate features:
 - debt
 - interest expense
-

Feature design example

```
operating_cash_flow_core = ocf + interest_paid  
free_cash_flow_core = operating_cash_flow_core - capex
```

```
leverage = debt / assets  
interest_burden = interest / revenue
```

-> This separates:

- operations
 - financing
-

7) When you WOULD include interest

Only if your target is:

- Equity returns
- Credit risk
- Default prediction

Then:

- interest matters directly

8) Clean mental model

- Remove interest → compare businesses
- Include interest → analyze investors (equity/debt)

☐ Final takeaway

Whether interest belongs in OCF depends on your goal: remove it to compare operating performance (FCFF-style), include it to model equity outcomes (FCFE-style). For ML, separating operations from financing is usually the safest choice.

If you want, I can help you design a full feature set (with formulas) tailored to your specific prediction task—that's the next high-leverage step.

Turn 45: User

Time: 2026-05-01 21:29:22 GMT+8

in the ifrs way of acct for development capex, is the amortization of it well described in the financial statement?

Turn 46: Assistant

Time: 2026-05-01 21:29:23 GMT+8

Short answer: yes—but not always in one clean line. Under IFRS (via IAS 38), companies must disclose amortization of intangible assets, including capitalized development—but you often have to piece it together.

1) Where you'll actually see it

A) Income statement (high-level)

You'll usually see something like:

- “Amortization of intangible assets”
- or bundled:
- “Depreciation and amortization”

-> This is total amortization, not just development

B) Notes to financial statements (this is key)

There is almost always a note like:

“Intangible assets”

Inside, you’ll typically find a roll-forward table:

Amount	
Opening balance	100
Additions (capitalized dev)	30
Amortization	(20)
Impairment	(5)
Closing balance	105

-> This is where:

- amortization is explicitly shown
- sometimes broken down by category (e.g., development vs licenses)

C) Cash flow statement (indirect clue)

- Amortization appears as:
- non-cash add-back in OCF

But:

- Not separated by type (dev vs other intangibles)

2) Is development amortization separately disclosed?

Sometimes yes, sometimes no

Best case:

You’ll see:

- “Amortization of capitalized development costs”

More common:

You get:

- Total intangible amortization
- PLUS a breakdown of intangible categories

-> Then you infer:

- Portion related to development
-

3) How analysts (and you) reconstruct it

You use the roll-forward:

$$\text{Amortization} = \text{Beginning} + \text{Additions} - \text{Ending} - \text{Impairments}$$

-> If breakdown is available:

- isolate development component
-

4) Required disclosures under IAS 38

Companies must disclose:

- Amortization methods
- Useful lives
- Gross carrying amount
- Accumulated amortization
- Reconciliation (roll-forward)

-> So the information exists, but:

not always neatly labeled for ML use

5) Practical reality (important for your pipeline)

Easy cases:

- European tech/pharma firms
- Often detailed breakdowns

Hard cases:

- Aggregated “intangibles”

- Development buried inside

-> You may need:

- footnote parsing
 - heuristics
-

6) What this means for your earlier adjustment

To normalize IFRS → GAAP:

You need:

- Capitalized development (from additions)
- Amortization of development

Then:

```
Adjusted earnings =  
  Reported earnings  
+ amortization (dev portion)  
- capitalized development
```

7) Key pitfall

☐ Don't double count

If you:

- subtract capitalized development
- but don't add back amortization

-> You're penalizing twice

8) Clean mental model

- Income statement → partial visibility
 - Cash flow → indirect signal
 - Notes → full truth (but messy)
-

☐ Final takeaway

Yes—IFRS requires disclosure of amortization of capitalized development costs, but it's often embedded within broader intangible asset disclosures, so you usually need to extract it from the footnotes rather than a single clean line item.

If you want, I can show a real 20-F example and how to parse it programmatically—that’s the step from theory → usable ML pipeline.

Turn 47: User

Time: 2026-05-01 21:39:33 GMT+8

ok show real 20-f ex

Turn 48: Assistant

Time: 2026-05-01 21:39:34 GMT+8

Great—let’s walk through a realistic 20-F–style example so you see exactly what you’d parse. I’ll use a structure very similar to what companies like SAP or Nokia disclose in their 20-F under IFRS.

□ 1) What a real 20-F note looks like

In the “Intangible assets” note, you’ll typically see a table like this:

Example (simplified from real filings)

€ millions	Development	Other intangibles	Total
-----	-----	-----	-----
Opening balance	500	800	1,300
Additions	200	100	300
Amortization	(150)	(120)	(270)
Impairment	(20)	(10)	(30)
FX / other	10	5	15
Closing balance	540	775	1,315

2) What you care about (for ML normalization)

From this table, you extract:

[OK] Capitalized development (THIS YEAR)

- Additions (Development) = 200

-> This is the amount IFRS capitalized instead of expensing

[OK] Amortization of development

- Amortization (Development) = 150

-> This is the portion hitting earnings this year

3) Where else it shows up

Income statement

You might see:

- "Amortization of intangible assets: 270"

-> But:

- Not split between development vs other
-

Cash flow statement

You'll see:

- "Capitalized development costs" (often in investing)

Example:

- CFI includes -200

-> This matches the "Additions" line

□ 4) How you reconstruct GAAP-like treatment

From the table:

- Capitalized = 200
 - Amortization = 150
-

Adjustment formula

```
Adjusted earnings =  
  Reported earnings  
+ amortization (150)  
- capitalized development (200)
```

-> Net effect:

- Earnings ↓ by 50
-

Balance sheet adjustment

Adjusted assets =
 Reported assets
 - development asset (540)

☐ 5) What if development is NOT separated?

Sometimes you get this instead:

€ millions	Total intangibles
-----	-----
Opening	1,300
Additions	300
Amortization	(270)

-> No breakdown ☐

Then you approximate:

Option A (best effort)

- Use:
- “capitalized development” line from cash flow (if disclosed)

Option B

- Assume a proportion based on:
 - prior years
 - industry norms
-

☐ 6) How you’d parse this programmatically

Step 1: locate the note

Search for:

- “intangible assets”
 - “development costs”
-

Step 2: extract rows

Look for keywords:

- “additions”

- “amortization”
 - “development”
-

Step 3: map to features

```
capitalized_dev = table["Additions"]["Development"]  
amortization_dev = table["Amortization"]["Development"]
```

□□ 7) Real-world quirks

Labels vary a lot

Instead of “development,” you might see:

- “internally generated software”
 - “product development”
 - “capitalized R&D”
-

Sometimes split across notes

- Additions in one table
 - Amortization buried elsewhere
-

8) Why this matters (tie back to your ML goal)

Without this step:

- IFRS company:
 - Higher earnings
 - Higher assets
- GAAP company:
 - Lower earnings
 - Lower assets

-> Model learns fake patterns

□ Final takeaway

In real 20-F filings, capitalized development and its amortization are usually disclosed in the “intangible assets” note via a roll-forward table—you extract “additions” and “amortization” for development and use them to rebuild

GAAP-like earnings.

If you want next, I can: -> show an actual raw excerpt (messy text) and how to parse it with regex / LLM extraction—that's the step from "structured table" → "real-world pipeline."

Turn 49: User

Time: 2026-05-01 22:27:58 GMT+8

can you explain the following

1. convert LIFO → FIFO using LIFO reserve. is lifo reservd detailed in 10-k?

2. adjusted_net_income =

net_income + impairment_losses

- impairment_reversals
- capitalized_development

do i also add back amortization from development capex?

3. Provisions (liabilities)

- IFRS → more flexible (principles-based)
- GAAP → stricter thresholds

-> IFRS may show:

- more liabilities earlier

Turn 50: Assistant

Time: 2026-05-01 22:27:59 GMT+8

Great set of questions—these are exactly the kinds of adjustments that matter in practice. I'll go one by one and keep it grounded.

1) Converting LIFO → FIFO using the LIFO reserve

What is LIFO vs FIFO?

- LIFO (Last-In, First-Out) → used under US GAAP
- FIFO (First-In, First-Out) → required under IFRS

-> IFRS does not allow LIFO

What is the LIFO reserve?

LIFO reserve = difference between FIFO inventory and LIFO inventory

$$\text{LIFO reserve} = \text{FIFO inventory} - \text{LIFO inventory}$$

Is it disclosed in the 10-K?

[OK] Yes (usually required under US GAAP)

Companies using LIFO must disclose either:

- "LIFO reserve" explicitly

OR

- "Excess of FIFO over LIFO cost"

-> Typically found in:

- Inventory footnote in the 10-K
-

Example

- Inventory (LIFO) = 80
- LIFO reserve = 20

-> FIFO inventory = 100

How to adjust

Balance sheet:

$$\text{FIFO inventory} = \text{LIFO inventory} + \text{LIFO reserve}$$

Income statement (important!)

Change in LIFO reserve affects COGS:

$$\text{FIFO COGS} = \text{LIFO COGS} - \Delta(\text{LIFO reserve})$$

-> Because:

- Increasing reserve → LIFO COGS is higher → adjust down
-

2) Your adjusted_net_income formula (key nuance)

You wrote:

```
adjusted_net_income =  
    net_income  
    + impairment_losses  
    - impairment_reversals  
    - capitalized_development
```

☐ Yes—you MUST add back amortization

Otherwise you double penalize

Correct version:

```
adjusted_net_income =  
    net_income  
    + impairment_losses  
    - impairment_reversals  
    - capitalized_development  
    + amortization_of_capitalized_development
```

Why?

Under IFRS:

- Year 1: no expense (capitalized)
- Future years: amortization expense

If you:

- subtract capitalization (correct)
- but don't add back amortization

-> you're counting the cost twice

Intuition

You're converting:

"spread over time"

into:

"expense immediately"

So:

- Remove future amortization
 - Replace with upfront expense
-

3) Provisions (this is subtle but important)

What is a provision?

A liability for a future uncertain obligation

Examples:

- Lawsuits
 - Warranties
 - Restructuring costs
-

Key difference

Under IFRS

- Recognize when:
- obligation is probable
- amount can be estimated

-> More judgment-based

Under US GAAP

- Typically requires:
- more certainty (often interpreted as “likely” with tighter thresholds)

-> More conservative / rules-based

Result

-> IFRS companies may:

- Recognize provisions earlier
 - Show:
 - higher liabilities
 - lower earnings (earlier)
-

Example

Lawsuit

- 60% chance of losing
 - Estimated loss: 100
-

IFRS:

- Record provision = 100
 - Expense now
-

GAAP:

- Might delay recognition
 - Or require higher certainty
-

Why this matters for your ML model

IFRS company may look:

- More conservative (lower earnings today)
- More leveraged (higher liabilities)

-> But it's timing, not necessarily economics

☐ Final takeaways

1) LIFO reserve

- Yes, usually disclosed in 10-K
 - Lets you convert to FIFO for comparability
-

2) Development adjustment

- Yes, add back amortization
 - Otherwise you double count
-

3) Provisions

- IFRS recognizes earlier (more judgment)

- GAAP delays until more certain

One-line mental model

Most IFRS vs GAAP differences are about timing (when to recognize), not whether something exists.

If you want, I can give you a “top 5 adjustments that actually move model performance”—so you don’t over-engineer everything.

Turn 51: User

Time: 2026-05-02 01:06:47 GMT+8

yes please give top 5.

btw just to clarify, when you record provisions in liabilities like expectation of lawsuit, you reduce the income so you reduce equity right?

Turn 52: Assistant

Time: 2026-05-02 01:06:47 GMT+8

Great—this is exactly the right level of focus. I’ll do both parts.

Top Top 5 adjustments that actually matter (for ML / analysis)

You don’t need 20 adjustments—these 5 get most of the signal.

1) Capitalized development (IFRS vs GAAP)

Why it matters: Huge distortion in:

- earnings
- assets
- ROA

Fix:

```
adjusted_earnings =  
    net_income  
    - capitalized_development  
    + amortization_of_development
```

-> Also remove development asset from balance sheet if you want full normalization

2) Impairments & reversals

Why it matters: Creates fake:

- losses (impairment)
- rebounds (reversals under IFRS)

Fix:

```
core_earnings =  
    net_income  
+ impairment_losses  
- impairment_reversals
```

3) Interest classification (cash flow)

Why it matters: IFRS vs GAAP classification differences distort OCF

Fix:

- Move all interest to financing

```
core_ocf = ocf + interest_paid
```

-> Makes operating performance comparable

4) Lease normalization (hidden debt)

Why it matters: Leases behave like:

- debt + asset

But accounting treatment varies

Fix:

- Include:
 - lease liabilities in debt
 - Treat lease expense consistently (optional advanced step)
-

5) LIFO → FIFO (inventory)

Why it matters: Affects:

- inventory
- COGS
- margins

Fix:

```
fifo_inventory = lifo_inventory + lifo_reserve  
fifo_cogs = lifo_cogs - change_in_lifo_reserve
```

Bonus (if you want a 6th)

6) Provisions (timing differences)

- IFRS → earlier recognition
- GAAP → later

-> Harder to adjust precisely, but:

- good to be aware for:
 - liabilities
 - earnings timing
-

☐ If you only do 3 things

Do these:

1. Development capitalization
2. Impairments
3. Interest normalization

-> That gets you ~80% of the benefit

[OK] Now your second question (important)

“when you record provisions... you reduce income so you reduce equity right?”

✓ ☐ Yes—exactly correct

Accounting flow

Step 1: Record provision

Example:

- Expected lawsuit loss = 100

Entry:

- Expense (income statement): -100

- Liability (balance sheet): +100

Step 2: Effect on equity

- Net income ↓ 100
- Retained earnings ↓ 100

-> So:

- Equity ↓ 100

Step 3: Balance sheet still balances

Assets	Liabilities + Equity
-----	-----
(unchanged)	+100 liability
	-100 equity

Intuition

A provision is:

“We expect to lose money in the future, so we recognize it now”

So:

- You don't wait for cash to leave
- You reduce earnings immediately

☐ When cash is actually paid later

- Liability ↓
- Cash ↓

-> No new impact on earnings (already recognized earlier)

☐ Clean mental model

- Provision = early recognition of future loss
- Effect:
- earnings ↓ now

- liability ↑
 - equity ↓
-

One-line takeaway

Yes—recording a provision reduces income and therefore reduces equity, because retained earnings reflect cumulative profits after recognizing expected losses.

If you want next, I can show how provisions can be manipulated to smooth earnings (“cookie jar reserves”)—that’s another classic Schilit-style insight.

Turn 53: User

Time: 2026-05-02 01:13:14 GMT+8

was it us gaap that accounted for interest on debt in ocf?

Turn 54: Assistant

Time: 2026-05-02 01:13:14 GMT+8

Yes — under US GAAP, interest paid is classified as operating cash flow (OCF).

[OK] US GAAP treatment

- Interest paid → OCF
- Interest received → OCF
- Dividends received → OCF
- Dividends paid → Financing (CFF)

-> No flexibility — it’s fixed

☐ ☐ Contrast with IFRS (important)

Under IFRS:

- Interest paid → Operating OR Financing
- Interest received → Operating OR Investing

-> Companies can choose (but must be consistent)

Why GAAP puts interest in OCF

GAAP views interest as:

a cost of running the business

So it sits with:

- salaries
- suppliers
- other operating expenses

☐ Why analysts often disagree

From an economic perspective:

- Interest depends on:
- how much debt you use
- Not on:
- how good your operations are

-> So analysts often:

- move interest to financing
- to isolate “pure operations”

☐ Final takeaway

Yes—US GAAP puts interest paid in operating cash flow, while IFRS allows flexibility, which is why analysts often normalize it when comparing companies.

Turn 55: User

Time: 2026-05-02 01:15:40 GMT+8

what are included in cash flow in financing? you said dividends and that is to shareholders?

Turn 56: Assistant

Time: 2026-05-02 01:15:41 GMT+8

Yes—financing cash flow (CFF) is all about how the company raises money from investors and returns money to them. And yes, dividends are payments to shareholders and show up in CFF.

Let's map it cleanly.

1) What “cash flow from financing” means

CFF = cash flows between the company and its capital providers

Capital providers =

- Debt holders (lenders)
- Equity holders (shareholders)

2) What is INCLUDED in financing cash flow

☐ Debt-related flows

- Borrowing money (new loans, bonds issued) → cash inflow
- Repaying principal → cash outflow

-> Example:

- Issue bond: +100
- Repay loan: -50

☐ Interest (sometimes)

- Under IFRS:
- Interest paid → can be financing
- Under US GAAP:
- Interest → operating (not financing)

☐ Equity-related flows

1) Issuing shares

- Selling stock to investors

→ cash inflow

2) Share buybacks (repurchases)

- Company buys its own stock

→ cash outflow

3) Dividends (your question)

-> Yes—dividends = payments to shareholders

- Always:
 - Financing cash outflow
-

☐ Lease principal (important modern item)

- Lease payments are split:
 - Principal portion → financing
 - Interest portion → operating or financing (depends)
-

3) What is NOT in financing

Just to avoid confusion:

- ☐ Operating expenses → OCF
 - ☐ Capex (buying assets) → investing (CFI)
 - ☐ Buying securities → investing
-

4) Example (full picture)

Company does:

- Borrow 100
 - Repay 40
 - Pay dividends 20
 - Buy back shares 30
-

Financing cash flow:

\[+100 - 40 - 20 - 30 = +10 \]

-> Net cash from financing = +10

5) Why this matters (tie to your earlier questions)

CFF tells you:

“How is this company funding itself?”

Interpretations

- Positive CFF
- Raising money (debt or equity)
- Negative CFF
- Returning money (dividends, buybacks, debt repayment)

6) Clean mental model

- OCF → operations
- CFI → investments
- CFF → financing (owners + lenders)

□ Final takeaway

Financing cash flow includes transactions with investors and lenders—borrowing, repaying debt, issuing stock, buybacks, and dividends (which are payments to shareholders).

If you want, I can show how companies manipulate CFF (e.g., buybacks vs dividends) to influence investor perception—that’s another subtle but important angle.

Turn 57: User

Time: 2026-05-02 07:56:35 GMT+8

India's IFRS profile says:

- Ind AS is based on and substantially converged with IFRS
- but India has not adopted IFRS for domestic companies
- Ind AS contains carve-outs and carve-ins

China's profile says:

- domestic listed companies use ASBE
- IFRS is not permitted for domestic companies in mainland China

can you explain carve outs/ins, and what is asbe?

Turn 58: Assistant

Time: 2026-05-02 07:56:35 GMT+8

Great question—this is exactly where “IFRS vs local GAAP” gets nuanced.

1) What are “carve-outs” and “carve-ins”?

When a country says:

“Our standards are based on IFRS”

...it doesn't always mean identical.

They may tweak IFRS → that's where these terms come in.

✂️ ☐ Carve-outs = removing or modifying parts of IFRS

“We don't agree with this IFRS rule → we change or remove it”

Example (conceptual)

IFRS might say:

- Recognize something at fair value

A country might:

- Require historical cost instead

→ That's a carve-out

☐ Carve-ins = adding extra rules on top of IFRS

“We want additional requirements beyond IFRS”

Example (conceptual)

- IFRS doesn't require a specific disclosure
- Local standard says:
- “You must disclose X”

-> That's a carve-in

Clean mental model

- Carve-out → IFRS rule removed or altered
 - Carve-in → extra rule added
-

2) What this means for India (Ind AS)

India uses:

Indian Accounting Standards

Which are:

- Based on IFRS
 - But not identical
-

Why India does this

- Align globally (IFRS-like)
 - But adapt for:
 - local regulation
 - economic conditions
 - policy goals
-

Implication (important for your ML thinking)

Ind AS $\approx$ IFRS, but:

Small differences can still distort comparability

3) What is ASBE (China)?

China uses:

Accounting Standards for Business Enterprises

What ASBE is

- China’s domestic accounting system
- Developed by Chinese regulators
- Substantially converged with IFRS, but not identical

Key point

From your note:

“IFRS is not permitted for domestic companies in mainland China”

-> So:

- Chinese listed companies:
- use ASBE, not IFRS
- Even if ASBE is “close” to IFRS

4) How similar is ASBE to IFRS?

- High-level → quite similar
- Details → can differ in:
- asset valuation
- government-related transactions
- disclosures

-> Think:

“IFRS-like, but sovereign-controlled”

5) Big picture comparison

Region	Standard	Relationship to IFRS
US	US GAAP	Separate system
EU / Korea	IFRS	Full adoption
India	Ind AS	IFRS + carve-outs/ins
China	ASBE	Converged but independent

6) Why this matters (your ML trap again)

You might have:

- IFRS company
- Ind AS company
- ASBE company

All in the same dataset

-> They look similar—but:

subtle rule differences → different earnings, assets, timing

7) Clean mental model

- IFRS adoption → identical rules
- Convergence (Ind AS, ASBE) → “close, but customized”
- Carve-outs/ins → where differences live

☐ Final takeaway

Carve-outs remove or modify IFRS rules, carve-ins add extra requirements, and systems like Ind AS (India) and ASBE (China) are IFRS-inspired but locally customized—so they’re similar, but not fully comparable.

If you want, I can show specific real carve-out examples (e.g., financial instruments or revenue differences in Ind AS vs IFRS)—that’s where the differences become concrete.

Turn 59: User

Time: 2026-05-02 12:26:04 GMT+8

yes specific examples of differences for ind as and asbe plz

Turn 60: Assistant

Time: 2026-05-02 12:26:05 GMT+8

Great—this is where “converged with IFRS” still hides real, model-breaking differences. I’ll give concrete examples for both India (Ind AS) and China (ASBE), focusing on things that actually move numbers.

India Ind AS vs IFRS — specific carve-outs / carve-ins

India’s Indian Accounting Standards is very close to IFRS, but there are targeted deviations.

1) Financial instruments — equity vs liability (big one)

IFRS (IAS 32 logic)

- Some instruments (e.g., certain preference shares) → liability

Ind AS carve-out

- More instruments allowed to be classified as equity
-

Example

Company issues redeemable preference shares:

- IFRS → liability (because redemption obligation)
 - Ind AS → may classify as equity
-

Impact

- Lower debt (Ind AS)
- Higher equity
- Better leverage ratios

-> Your model might think:

“This company is less leveraged”

...but it's classification

2) Foreign currency convertible bonds (FCCBs)

IFRS

- Conversion option often treated as a derivative liability
- Measured at fair value → P&L volatility

Ind AS carve-out

- Can classify conversion option as equity
-

Impact

- IFRS → volatile earnings
- Ind AS → smoother earnings

-> Huge for ML signals around volatility

3) Business combinations under common control

IFRS

- No single prescribed method (often book-value or acquisition depending on interpretation)

Ind AS carve-in

- Pooling of interests method required
-

Impact

- No goodwill recognized
- No revaluation

-> Assets and earnings differ vs IFRS treatment

4) Investment property measurement

IFRS

- Option:
- fair value model (no depreciation)
- or cost model

Ind AS carve-out

- Fair value model NOT allowed for measurement
 - Only cost model (fair value disclosed separately)
-

Impact

- Lower asset volatility
 - Lower reported asset values vs IFRS fair value users
-

China ASBE vs IFRS — specific differences

China's Accounting Standards for Business Enterprises is also IFRS-like, but with policy-driven differences.

1) Government grants (very important in China)

IFRS

- Recognize when:
- “reasonable assurance” of receipt and conditions met

ASBE

- Often more structured / prescriptive
 - Closer alignment with government policy timing
-

Example

Subsidy for manufacturing:

- IFRS → recognized when conditions reasonably assured
 - ASBE → may align with formal approval or receipt timing
-

Impact

- Timing differences in income
 - Earnings comparability issues
-

2) Related-party disclosures (state-owned enterprises)

IFRS

- Requires extensive disclosure of related parties

ASBE carve-out

- Reduced disclosure requirements for state-controlled entities
-

Impact

- Less transparency
- Harder to detect:
- hidden transactions
- risk exposures

-> Important for ML anomaly detection

3) Asset impairment reversals (more restrictive in practice)

IFRS

- Allows reversal (except goodwill)

ASBE

- In practice, more conservative / restrictive application
-

Impact

- Less earnings “bounce back” vs IFRS
 - Different time-series patterns
-

4) Fair value usage

IFRS

- Broad use of fair value across assets

ASBE

- More cautious / limited use in some areas
-

Impact

- Less volatility
 - More conservative asset values
-

5) Development costs (subtle but real)

IFRS

- Capitalize if criteria met

ASBE

- Similar rules, but:
 - often more conservative in practice
-

Impact

- Less capitalization → lower assets
- Lower earnings vs IFRS-heavy capitalizers

Cross-country intuition (very important)

Area	IFRS	Ind AS	ASBE
Philosophy	Principles	IFRS + tweaks	IFRS-like + policy
Equity vs liability	Strict	More equity classification	Similar to IFRS
Fair value	Broad	Restricted in places	More conservative
Gov influence	Low	Medium	High

What actually matters for your ML model

Focus on differences that change:

1. Capital structure

- Equity vs liability (Ind AS)

2. Earnings volatility

- Derivatives / fair value (Ind AS vs IFRS)

3. Timing of income

- Government grants (ASBE)

4. Transparency

- Related-party disclosures (ASBE)

□ Final takeaway

Ind AS and ASBE are IFRS-inspired but contain targeted carve-outs (removals/changes) and carve-ins (additions) that affect classification, timing, and disclosure—especially for financial instruments, government involvement, and fair value—so they can materially distort cross-country comparisons and ML features.

If you want, next step I'd suggest is: -> a "minimal normalization layer" specifically for India/China datasets (what to fix vs what to ignore)

Turn 61: User

Time: 2026-05-02 15:27:03 GMT+8

how do you know which acct standard each company use? is it in annual report? how about 10k and 20f?

Turn 62: Assistant

Time: 2026-05-02 15:27:04 GMT+8

Yes—you can reliably identify a company's accounting standard from its filings, and for U.S.-listed companies the form type alone already gives you a strong signal.

Let's break it down cleanly.

1) Where to find the accounting standard (most reliable places)

A) Annual report / Form 10-K / Form 20-F

Every company must explicitly state something like:

- "Prepared in accordance with US GAAP"
 - or
 - "Prepared in accordance with **IFRS as issued by the IASB"
-

☐ Where exactly in the report?

Look for:

1. "Basis of preparation" section (best place)

Usually in:

- Note 1 to financial statements
 - "Summary of significant accounting policies"
-

2. Auditor's report

Often says:

- "In conformity with US GAAP"
 - or "in accordance with IFRS"
-

3. Footnotes

Same language repeated

2) 10-K vs 20-F (very useful shortcut)

☐ Form 10-K

Filed by U.S. domestic issuers

-> Almost always:

- US GAAP
-

☐ Form 20-F

Filed by foreign private issuers

-> Usually:

- IFRS

But sometimes:

- Local GAAP (less common now)
-

☐ Quick heuristic

Filing Likely standard
----- -----
10-K US GAAP
20-F IFRS (usually)

3) Non-U.S. companies (outside SEC filings)

If you're using global data:

☐☐ Europe / ☐☐ Korea / many others

- Use IFRS (mandatory for listed firms)
-

India India

- Uses Indian Accounting Standards
-

China China

- Uses Accounting Standards for Business Enterprises
-

-> Found in:

- Annual report
 - “Accounting policies” section
-

4) Programmatic detection (for your ML pipeline)

Option A — Filing type (fast heuristic)

```
if filing_type == "10-K":  
    standard = "US_GAAP"  
elif filing_type == "20-F":  
    standard = "IFRS"
```

Option B — Text parsing (more robust)

Search for keywords:

- “US GAAP”
 - “IFRS as issued by the IASB”
 - “Ind AS”
 - “ASBE”
-

Option C — Data vendor fields

If using:

- Bloomberg
- FactSet
- Refinitiv

-> They usually have:

- accounting_standard field
-

5) Edge cases to watch

☐ ☐ 1. Dual listings

- Same company listed in multiple markets
 - Still uses one accounting standard
-

☐ ☐ 2. IFRS vs “IFRS as issued by IASB”

- Important distinction for SEC acceptance
-

☐ 3. Historical transitions

- Company may switch standards over time

-> Your model should:

- track standard by year

6) Best practice for your ML pipeline

Step 1:

Create feature:

```
accounting_standard {
  US_GAAP,
  IFRS,
  IND_AS,
  ASBE,
  OTHER
}
```

Step 2:

Validate using:

- filing type
- text parsing

Step 3:

Store per company-year

Clean mental model

- 10-K → US GAAP
- 20-F → usually IFRS
- Annual report → always explicitly states it

☐ Final takeaway

Yes—the accounting standard is explicitly disclosed in annual reports (especially in the “basis of preparation” note), and for U.S.-listed firms you can often infer it quickly: 10-K = US GAAP, 20-F = usually IFRS.

If you want, I can show how to build a robust parser that extracts accounting standard from raw filings (including messy real text)—that’s very useful for automation.

Turn 63: User

Time: 2026-05-02 19:30:28 GMT+8

Under IFRS 17, effective January 1, 2023:

- insurance cash flows are measured on a current basis
- profit is recognized over service periods
- insurance service result is presented separately from insurance finance income or

expense That is a completely different economic structure from a normal operating company. So: My current three-statement operating-company framework is not the right native model for insurers. I would treat insurance as a separate product line with a distinct state representation and accounting engine.

so the insurance service does not appear in balance sheet, income statement, or cash flow??

Turn 64: Assistant

Time: 2026-05-02 19:30:29 GMT+8

Short answer: **it absolutely does appear in the financial statements**—just not in the same way you’re used to with a “normal” operating company.

The quote you’re reading is saying:

the structure is different, not that it’s “outside” the statements.

Let’s map it cleanly.

1) Where “insurance service” shows up

Under IFRS 17:

Income statement (primary place)

You’ll see a split like:

A) Insurance service result

- Revenue from insurance contracts
- Claims and expenses
- Release of profit (CSM—more on that below)

-> This is the core operating performance

B) Insurance finance income/expense

- Effects of discount rates
- Time value of money

-> More like “financial” noise

☐ So:

“Insurance service result” = operating result for insurers

2) Where it shows up in the balance sheet

Even though “insurance service” isn’t a line item, it’s embedded in liabilities:

Insurance contract liabilities include:

- Expected future cash flows
 - Risk adjustment
 - Contractual Service Margin (CSM)
-

What is CSM?

Deferred profit

- Profit you expect to earn in the future
 - Recognized over time as service is provided
-

Key link:

- CSM sits on balance sheet (liability)
 - As service is delivered:
 - CSM ↓
 - Profit ↑ (in income statement)
-

3) Cash flow statement

Cash flows still exist, but:

- Premiums received → operating inflow

- Claims paid → operating outflow

-> However:

Cash flow is less aligned with profit than in normal companies

Because:

- Profit is smoothed via CSM
 - Cash can be volatile
-

4) Why this feels “different”

In a normal company:

- Revenue $\approx$ cash + accrual adjustments
 - Costs recognized when incurred
-

In insurance:

- You collect cash upfront (premiums)
- But recognize profit over time

-> So:

Cash $\neq$ profit timing

5) Concrete example

Customer pays \$100 premium today for 1-year coverage

Day 1:

- Cash $\uparrow$ 100
 - Liability $\uparrow$ (future obligation)
 - Profit $\approx$ 0
-

Over the year:

- You provide insurance service
- Gradually recognize profit

-> Through:

- release of CSM

6) Why the quote says your model breaks

A standard 3-statement model assumes:

- Revenue ↔ production ↔ cost

But insurers operate like:

“liability engines” managing risk over time

Key state variables:

- Expected claims
- Discount rates
- Risk adjustment
- CSM (deferred profit)

7) Clean mapping to your framework

Concept	Normal company	Insurance (IFRS 17)
-----	-----	-----
Revenue	Sales	Insurance service revenue
Profit timing	Near-term	Spread via CSM
Balance sheet	Assets	Large liabilities (CSM etc.)
Cash vs profit	Aligned-ish	Decoupled

8) Final clarification (your question)

“does insurance service not appear in statements?”

☐ Not correct

It does appear, but:

- Not as a balance sheet line item
- Appears as:
 - income statement component
 - movement in liabilities (CSM)

☐ Final takeaway

Insurance service under IFRS 17 is very much in the financial statements—it shows up as “insurance service result” in the income statement and through changes in insurance liabilities (especially CSM) on the balance sheet—but its timing and structure are fundamentally different from a typical operating company.

If you want, I can show a real insurer income statement under IFRS 17 (e.g., Allianz or AXA) and map each line to your model—that’s where it really clicks.

Turn 65: User

Time: 2026-05-02 19:33:58 GMT+8

csm covers claims?

Turn 66: Assistant

Time: 2026-05-02 19:33:59 GMT+8

Short answer: ****No—CSM does not cover claims. It represents future profit**, not expected costs.**

Let’s make this precise.

1) What CSM actually is

Under IFRS 17:

CSM (Contractual Service Margin) = unearned profit

It’s the portion of the contract’s value that:

- Cannot be recognized as profit yet
- Will be released over time as insurance service is provided

2) What does cover claims?

Claims are covered by a different component:

☐ Fulfilment cash flows

These include:

- Expected future claims
- Expenses

- Discounting (time value)
 - Risk adjustment
-

3) Full structure of insurance liability

Think of it like:

```
Insurance liability =  
  Expected claims & expenses (fulfilment cash flows)  
+ Risk adjustment  
+ CSM (profit)
```

Key distinction

Component	What it represents
----- -----	
Fulfilment cash flows	Costs (claims, expenses)
Risk adjustment	Uncertainty buffer
CSM	Profit

4) Example (very intuitive)

Customer pays premium = 100

Expected:

- Claims = 70
- Expenses = 10

So:

- Total cost = 80
 - Profit = 20
-

At inception:

- Liability includes:
- 80 (costs)
- 20 (CSM)

-> No profit recognized yet

5) Over time

As insurance service is provided:

- Claims happen → reduce fulfilment cash flows
 - CSM is gradually released → becomes profit
-

Important

- If claims increase unexpectedly:
 - CSM may shrink
 - If losses exceed expectations:
 - CSM can go to zero
 - Loss hits earnings
-

6) Why this matters

If you confuse CSM with claims:

You might think:

“CSM is a reserve for payouts”

But actually:

****CSM is the opposite—it’s the profit you expect after paying claims****

7) Clean mental model

- Claims → costs → fulfilment cash flows
 - CSM → profit → deferred and released over time
-

☐ Final takeaway

CSM does not cover claims—it represents the insurer’s expected future profit after accounting for claims and expenses, and it is recognized gradually as insurance service is delivered.

If you want, I can walk through a full lifecycle example (premium → claims → CSM release → income statement) step-by-step—that’s the best way to internalize it.

Turn 67: User

Time: 2026-05-02 19:35:13 GMT+8

is csm like deferred revenue in apples balance sheet? which i think is like apple care?

Turn 68: Assistant

Time: 2026-05-02 19:35:13 GMT+8

You're thinking in exactly the right direction — **CSM is similar to deferred revenue, but they're not the same thing**. The similarity is useful; the differences matter.

1) The intuition you have is mostly right

- Apple's deferred revenue (e.g., AppleCare)
- Insurer's CSM

-> Both represent:

"Cash received (or value locked in) but profit not yet recognized"

☐ 2) Apple deferred revenue (your example)

At Apple Inc.:

Example: AppleCare

- Customer pays upfront (say \$200)
 - Apple provides service over time
-

At purchase:

- Cash ↑
 - Deferred revenue (liability) ↑
-

Over time:

- Revenue recognized gradually
- Liability ↓

-> This is straightforward "unearned revenue"

□□ 3) CSM (insurance)

Under IFRS 17:

CSM =

Expected future profit, deferred and released over time

At inception:

- Premium received
- Expected claims + costs estimated
- Remaining profit = CSM

Over time:

- CSM ↓
- Profit recognized

□□ 4) Key differences (this is the important part)

Difference 1 — What they represent

		Deferred revenue		CSM	
	--	-----		-----	
	Meaning		Unearned revenue		**Unearned profit**

-> Big distinction:

- Apple hasn't delivered service → revenue not earned
- Insurer has priced contract → profit known but deferred

Difference 2 — Risk

Apple:

- Cost uncertainty relatively low

Insurance:

- Claims are uncertain

-> So CSM depends on:

- expected claims
- risk adjustments
- discount rates

Difference 3 — Measurement complexity

Deferred revenue:

- Based on cash received

CSM:

- Based on:
- actuarial estimates
- future cash flows
- risk adjustments

-> Much more model-driven

Difference 4 — Volatility

Deferred revenue:

- Smooth and predictable

CSM:

- Can change if:
- claims expectations change
- assumptions update

☐ 5) Where your analogy works well

Think of:

CSM ≈ “deferred profit”
Deferred revenue ≈ “deferred revenue”

Better analogy

If AppleCare were accounted like insurance:

- Apple would estimate:

- future repair costs
- Separate:
- cost portion
- profit portion

-> Then defer only the profit → that would look like CSM

☐ Final takeaway

CSM is conceptually similar to deferred revenue (like AppleCare), but instead of deferring revenue, it defers expected profit after accounting for future claims and costs—making it more complex and model-driven.

If you want, I can show a side-by-side accounting flow (AppleCare vs insurance policy)—that comparison really locks it in.

Turn 69: User

Time: 2026-05-02 19:38:51 GMT+8

whats discount rate

Turn 70: Assistant

Time: 2026-05-02 19:38:51 GMT+8

A discount rate is the rate you use to convert future cash flows into today's value.

1) Core idea (intuitive)

Money in the future is worth less than money today

Why?

- You could invest today and earn returns
- There's uncertainty (risk)
- Inflation reduces purchasing power

2) The formula

$$PV = \frac{CF}{(1+r)^t}$$

- PV = present value

- CF = future cash flow
 - r = discount rate
 - t = time
-

3) Simple example

You expect \$100 in 1 year.

- Discount rate = 10%

$$[PV = 100 / 1.1 = 90.9]$$

-> That \$100 in the future is worth ~\$91 today

4) What determines the discount rate?

Depends on context, but generally includes:

1. Risk-free rate

- Government bond yield

2. Risk premium

- Extra return for uncertainty
-

Example

- Risk-free = 3%
- Risk premium = 7%

-> Discount rate = 10%

5) Why discount rate matters A LOT

Small changes → big effects

- Higher discount rate → lower present value
 - Lower discount rate → higher present value
-

6) Where you've already seen it (tie to your questions)

A) Insurance (IFRS 17)

Under IFRS 17:

- Future claims are discounted
 - Discount rate affects:
 - liabilities
 - profit timing
-

B) Valuation (DCF)

- Used to value companies
 - Determines:
 - how much future cash is worth today
-

C) Accounting (provisions, leases)

- Future obligations → discounted
 - Affects:
 - liabilities
 - expenses
-

7) Clean mental model

- Discount rate = “how impatient and risk-aware you are”

Higher rate:

“I value future money much less”

Lower rate:

“Future money is almost as good as today”

☐ Final takeaway

The discount rate is the rate used to translate future cash flows into present value, reflecting time, risk, and opportunity cost—and it plays a critical role in valuation, insurance accounting, and financial reporting.

If you want, I can show how changing discount rates affects insurer earnings under IFRS 17—that’s a really powerful (and non-obvious) effect.

Turn 71: User

Time: 2026-05-02 20:59:55 GMT+8

if i were to adapt my financial model to lets say banks, how can i account for its difference from tech firms namely deposits and loans and loss reserve etc? would i model it similar to how i modeled liquidity of a firm as a ratio of revenue?

also, tech firms are apparently heavy in nontangibles. can you explain?

Turn 72: Assistant

Time: 2026-05-02 20:59:55 GMT+8

You'll get much better results if you don't try to force banks into a "revenue-cost-capex" tech template. Banks are closer to balance-sheet (spread) machines than product companies.

Let's split your question into two parts.

☐ 1) How to model banks (vs tech firms)

☐ What not to do

Don't model banks like:

- revenue → expenses → EBITDA → capex

That works for tech, not banks.

[OK] Core idea for banks

Banks generate profit from the spread between assets and liabilities

☐ Key building blocks

1) Balance sheet is the engine

Instead of focusing on revenue, focus on:

- Loans (assets)
 - Deposits (liabilities)
-

2) Net Interest Income (NII)

Core “revenue” for a bank

$$\text{NII} = \text{Interest on loans} - \text{Interest on deposits}$$

3) Net Interest Margin (NIM)

$$\text{NIM} = \frac{\text{NII}}{\text{Interest-earning assets}}$$

-> This is like “gross margin” for banks

4) Credit losses (VERY important)

Instead of “COGS”, you have:

- Provision for loan losses

This is:

- expected defaults
- modeled via loss reserves

5) Loan loss reserve (balance sheet)

Cushion for expected loan defaults

- Expense → increases reserve
- Actual write-offs → reduce reserve

How to structure your model

Instead of revenue-based:

Step 1: Model balance sheet

- Loans growth
- Deposits growth
- Capital

Step 2: Generate income

Interest income = loans × yield
Interest expense = deposits × cost
NII = difference

Step 3: Add credit cost

Provision = % of loans (expected loss)

Step 4: Add operating cost

- Salaries
 - branches
 - tech
-

Step 5: Profit

Net income =
NII
- provisions
- operating expenses

☐ Key insight

Deposits are NOT like revenue—they are funding

So your idea:

“model liquidity as ratio of revenue”

-> ☐ Not appropriate for banks

[OK] Better ratios

Use:

- Loan-to-deposit ratio (LDR)
 - Capital adequacy (CET1 ratio)
 - Non-performing loans (NPL ratio)
 - Coverage ratio (reserves / bad loans)
-

Clean mental model

- Tech firm → income statement driven
 - Bank → balance sheet driven
-

□ 2) Why tech firms are “heavy in intangibles”

This is a big conceptual shift.

What are intangible assets?

Under IAS 38:

- Software
 - patents
 - algorithms
 - brand
 - data
-

Why tech is intangible-heavy

1) Value comes from ideas, not machines

- Code
- network effects
- IP

-> Not factories or equipment

2) Accounting mismatch

Here's the tricky part:

-> Many intangibles are NOT on the balance sheet

Example

A company spends on:

- engineers
- product development

Under US GAAP:

- Expensed → no asset recorded
-

Result

- Real value exists
- Accounting assets look small

-> ROA looks artificially high

3) When intangibles do appear

- Acquisitions → goodwill
- IFRS development capitalization

4) Implication for modeling

For tech firms:

- Balance sheet understates true assets
- Earnings may be suppressed (expensed R&D)

Better metrics

- Revenue growth
- Gross margin
- R&D intensity
- Cash flow

□ Big contrast (this is the key takeaway)

		Tech firm		Bank	
	--	-----		-----	
	Core driver		Product / IP		Balance sheet spread
	Assets		Hidden (intangibles)		Explicit (loans)
	Revenue		Sales		Interest spread
	Risk		Market / product		Credit risk
	Model type		Income statement		Balance sheet

□ Final takeaway

Banks should be modeled as balance-sheet-driven spread businesses (loans, deposits, credit losses), not revenue-driven firms, while tech companies are “intangible-heavy” because their value comes from IP and R&D that is often not fully captured on the balance sheet.

If you want, I can show: -> a minimal bank model (5–6 equations) you can plug directly into your ML or simulation framework

Turn 73: User

Time: 2026-05-03 15:23:34 GMT+8

Ongoing taxonomy and standards maintenance Missing:

- no formal update process for IFRS, US GAAP taxonomy, ESEF, or SEC filing changes

Needed:

- standards monitoring
- regression suite against taxonomy changes
- controlled rollout by reporting season

This is especially important because:

- IFRS 18 becomes effective January 1, 2027
- ESMA published a 2025 ESEF taxonomy on April 21, 2026 with IAS 1 and IFRS 18 entry

points for 2026 reporting preparation

what does it mean by regression?

Turn 74: Assistant

Time: 2026-05-03 15:23:35 GMT+8

Great question—here “regression” has nothing to do with statistical regression models. It means regression testing.

1) What “regression” means here

Regression testing = checking that existing functionality still works after a change

In your context:

- Accounting standards change (e.g., IFRS 18)
- Taxonomies update (ESEF, SEC XBRL tags)

-> You need to ensure:

“Did anything break in our pipeline?”

☐ 2) What is being tested?

Your system likely does things like:

- Parse financial statements
 - Map line items to taxonomy tags
 - Compute features (OCF, EBITDA, etc.)
-

After a taxonomy update:

- Tag names may change
- Structure may change
- New line items added

-> Risk:

- Your parser fails
 - Features become wrong
 - Models degrade silently
-

☐ 3) What a “regression suite” looks like

You maintain a set of test cases:

Example:

- Apple 10-K (US GAAP)
 - SAP 20-F (IFRS)
 - Some ESEF filings
-

Tests check:

- “Revenue extracted = correct?”
 - “OCF matches known value?”
 - “Tags mapped correctly?”
-

After update:

You run tests again:

- If outputs change unexpectedly → ☐ regression failure

- If unchanged (or expected change) → [OK] pass
-

4) Concrete example

Before taxonomy update

- Tag: Revenue
 - Your system maps correctly
-

After update (e.g., European Securities and Markets Authority ESEF update)

- Tag renamed or restructured

-> Without regression testing:

- Your system may return null or wrong value
-

☐ 5) Why this is critical (your context)

You mentioned:

- IFRS 18 coming
- New ESEF taxonomy

These can:

- Change presentation structure
- Introduce new categories
- Modify tagging rules

-> That's exactly when regressions happen

☐ 6) What “controlled rollout by reporting season” means

Instead of:

flipping everything at once

You:

1. Run regression tests
2. Validate outputs
3. Gradually deploy

-> Especially around:

- Q1 / annual reporting periods
-

7) Clean mental model

- Standards/taxonomy update = code change to your data layer
 - Regression testing = unit tests for your financial pipeline
-

☐ Final takeaway

“Regression” here means regression testing—running a suite of checks to ensure that updates to accounting standards or taxonomies (like IFRS 18 or ESEF changes) don’t break your data extraction, mappings, or model inputs.

If you want, I can show how to design a minimal regression test suite specifically for financial statement parsing—that’s super useful for your pipeline.

Turn 75: User

Time: 2026-05-05 09:46:40 GMT+8

Can you generate a pdf file of this conversation?

Turn 76: Assistant

Time: 2026-05-05 09:46:56 GMT+8

I’ll create a clean, downloadable PDF transcript of the visible conversation with speaker labels and readable section breaks, excluding internal system or developer instructions.

Turn 77: Assistant

Time: 2026-05-05 09:49:59 GMT+8

I’m formatting it more like structured study notes than a raw chat dump, while preserving the sequence and substance of the accounting and ML discussion.

Turn 78: Assistant

Time: 2026-05-05 09:51:46 GMT+8

Done — I generated the PDF as structured transcript-style notes from the conversation.

[Download the PDF](sandbox:/mnt/data/accounting_ml_conversation_notes.pdf)

Turn 79: User

Time: 2026-05-05 11:02:57 GMT+8

i cant download the file. can you generate the link again?

Turn 80: Assistant

Time: 2026-05-05 11:03:13 GMT+8

I'll verify the PDF still exists in the workspace and then give you a fresh sandbox link.

Turn 81: Assistant

Time: 2026-05-05 11:06:15 GMT+8

The old file is gone, so I'm recreating it under a new filename. A minor ReportLab style-name issue came up, and I'm rerunning with a corrected style.

Turn 82: Assistant

Time: 2026-05-05 11:07:51 GMT+8

Here's a fresh regenerated link:

[Download the PDF notes](sandbox:/mnt/data/accounting_ml_conversation_notes_v2.pdf)